

Post-translational modifications of microtubules are crucial for malaria parasite transmission

Reviewed Preprint

v1 • August 27, 2025

Not revised

Kodzo Atchou , Magali Roques, Ruth Rehmann, Reto Caldelari, Melanie Schmid, Simone Grossi, Bianca Manuela Berger, Torsten Ochsenreiter, Friedrich Frischknecht, Volker Heussler 

Institute of Cell Biology, University of Bern, Bern, Switzerland • Graduate School for Cellular and Biomedical Sciences, University of Bern, Bern, Switzerland • Department of Parasitology, University of Heidelberg Medical School, Heidelberg, Germany

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study provides **valuable** insights into microtubule remodeling during liver-stage *Plasmodium berghei* development, demonstrating that deletion of the alpha-tubulin C-terminal tail impairs parasite growth in mosquitoes and abolishes infection in HeLa cells. The work is technically ambitious, employing advanced microscopy, genetic mutants, and pharmacological approaches. However, key claims are only partially supported due to **incomplete** evidence linking tubulin modifications to microtubule dynamics and uncertain antibody-based PTM detection.

<https://doi.org/10.7554/eLife.108209.1.sa4>

Abstract

Microtubules, composed of α - and β -tubulin polymers, are essential components of the eukaryotic cytoskeleton. They maintain cellular shape and structural integrity and play critical roles in cell division and in intracellular vesicular transport. In *Plasmodium*, the parasite that causes malaria, nuclear replication during the liver stage is among the fastest known in eukaryotic cells and relies heavily on microtubules for DNA segregation and cytoskeletal organization. Despite their importance, the role of microtubules in liver stage development remains largely unexplored. Here, we investigated microtubule dynamics during liver stage development using a combination of cell and molecular biology techniques, expansion microscopy, and live-cell imaging. By employing antibodies specific for α -tubulin post-translational modifications (PTMs), we found that the *Plasmodium* sporozoites subpellicular microtubules (SSPM) persist during liver infection, giving rise to liver stage parasite microtubule bundles (LSPMB). These LSPMB form multimeric tubulin structures within hepatocytes and are redistributed to the hemi-spindle poles of parasite nuclei during schizogony. Deletion of the C-terminal region encompassing all known *Plasmodium* α -tubulin PTM sites prevented sporozoite migration from the mosquito midgut to the salivary glands, effectively blocking parasite transmission. Using *Plasmodium* microtubule-specific depolymerisation drugs, we found that while LSPMB are stable in sporozoites, they exhibit dynamic behavior during hepatocyte infection. Given the regulatory role of PTMs in

microtubule dynamics, we generated parasite mutants by substituting and deleting key α -tubulin C-terminal residues involved in PTMs. Substitution of the polyglutamylation site with alanine and deletion of the C-terminal tyrosination/detyrosination motifs impaired parasite growth during liver infection. Together, our findings reveal extensive microtubule remodeling during liver stage development and establish α -tubulin C-terminal modifications as critical regulators of both intracellular development and parasite transmission of *Plasmodium* parasites.

Introduction

Plasmodium parasites are responsible for malaria and cause yearly half a million deaths in sub-Saharan countries. Malaria continues to be a major worldwide health concern despite recent successful attempts to lower its prevalence (1–4). The infection starts when an infected female *Anopheles* mosquito, during a blood meal, injects salivary gland sporozoites into the skin of the vertebrate host. Some of the motile sporozoites escape the host immune defence and invade blood vessels. They are carried by the blood flow to the liver, where they invade and infect hepatocytes (5). Approximately 50% of the parasites are often killed by host cell responses (6–8). Those that managed to establish a productive invasion form a parasitophorous vacuole (PV) in which the parasite undergoes massive growth and asexual replication known as schizogony. One single parasite gives rise to thousands of daughter parasites called merozoites in 3 to 6 days depending on the *Plasmodium* species (9–11). Fully developed merozoites exit the liver cells in merosome vesicles through adjacent blood vessels (11–14). The merosomes then rupture and release merozoites into the blood stream. The parasites immediately invade erythrocytes, where they undergo again asexual replication to form the blood stage merozoites. Some merozoites differentiate into sexual stages, termed gametocytes, which are ingested by female mosquitos during a blood meal. In the mosquito midgut, the gametocytes develop into microgametes and macrogametes, which fertilize and form a zygote. The later undergoes differentiation into motile ookinetes, which penetrate the gut wall, forming oocysts (15–18). The oocyst produces the midgut sporozoites from the sporoblasts. The sporozoites migrate to the salivary gland from day 11 post-infection in *P. berghei* (Frischknecht et al., 2004; Frischknecht & Matuschewski, 2017; Jerome P. Vanderberg, 1974; J. P. Vanderberg, 1975). During the liver stage development, the parasite relies on the fast and massive asexual replication which is essential for the transmission and pathogenesis in erythrocyte stage, representing key targets for malaria control strategies. Investigating the parasite microtubule dynamics during the rapid replication could provide important insight on the parasite biology and to develop strategies for halting the massive proliferation during the clinically asymptomatic liver stage.

Microtubules are cytoskeletal filaments of about 25nm outer diameter, made up of usually 13 chains of protofilaments composed of globular α -tubulin and β -tubulin heterodimers. They are crucial components of the cytoskeleton and offer structural support to cells and also function as tracks for the movement of organelles, vesicles, and other cellular components (23,24). Microtubules are involved in important cellular activities such as cell division and the production of cilia and flagella to allow cell motility. Microtubules exhibit inherent dynamics, alternating between polymerisation (rapid growth) and spontaneous depolymerisation (shrinkage) in a process termed dynamic instability (23,25,26). The dynamics of microtubules and their interactions with other cellular structures are regulated by specific microtubule-associated proteins (MAPs), which selectively bind to either the plus or minus ends of microtubules (27–30).

Tubulin is subject to seven identified post-translational modifications (PTMs), among which are acetylation of the lysine residue at position 40 in the lumenally exposed part of α -tubulin, glycylation, which entails the incorporation of polyglycine side chains of variable length, phosphorylation, palmitoylation, tyrosination/ detyrosination and polyglutamylation. These

modifications are extensively preserved and enabling cells to modulate microtubule characteristics, dynamics, and interactions with microtubule-associated proteins (MAPs) and motor proteins (31 [↗](#), 32 [↗](#)). Tyrosinated microtubules are often associated with microtubule stability. These stable microtubules represent a specialised subset that exhibits enhanced resistance to depolymerization, even in the presence of destabilizing agents. They play a crucial role in maintaining cellular architecture, facilitating intracellular transport, and supporting specialized cellular functions. In contrast to dynamic microtubules, which continuously undergo cycles of polymerization and depolymerization due to dynamic instability, stable microtubules persist for extended periods. They are also frequently polyglutamylated, a modification that regulates their interactions with motor proteins and other cellular components. These microtubules are predominantly localized in structures requiring sustained structural integrity, such as cilia, flagella, neuronal axons, and motile cell types (33 [↗](#)–36 [↗](#)). Glutamylation, or polyglutamylation, is a post-translational modification characterized by the addition of multiple glutamate side chains to glutamate residues on both α - and β -tubulin. This modification occurs primarily at the C-terminal tails and is catalyzed by a family of enzymes known as polyglutamylases. Polyglutamylation plays a critical role in regulating the interactions of microtubules with MAPs and motor proteins, thereby influencing microtubule dynamics and cellular functions (37 [↗](#)–40 [↗](#)). The properties and functions of tubulin are largely encoded within their protein sequences, contributing to molecular diversity that influences key microtubule characteristics. This variation has been conceptualized as the “tubulin code”, a regulatory framework through which cells control microtubule diversity to achieve specific physiological functions (24 [↗](#)). Nevertheless, the precise mechanisms by which cells establish and interpret this code remain largely unknown. Although tubulin is highly conserved across evolution, it exhibits structural and functional variations between species, cell types, and even within single cells.

In eukaryotic cells, tubulins are key components of the microtubule cytoskeleton, and their genetic complexity reflects the organism’s level of cellular specialization. In yeast, two distinct genes encode α -tubulin, while a single gene encodes β -tubulin. In contrast, the human genome contains nine genes encoding α -tubulin and an equal number encoding β -tubulin, reflecting a greater level of complexity and functional specialization in higher eukaryotes (24 [↗](#), 41 [↗](#), 42 [↗](#)).

The *Plasmodium* genome encodes two α -tubulins (α 1-tubulin and α 2-tubulin) and one β -tubulin (43 [↗](#)–47 [↗](#)). The α 1-tubulin gene encodes a protein of 450 amino acids, while α 2-tubulin encodes 453 amino acids (48 [↗](#)). The two isoforms share 94% amino acid identity. However, α 2-tubulin lacks a terminal tyrosine residue, which is present in α 1-tubulin. This difference suggests potential variation in post-translational modifications and functional roles (44 [↗](#), 49 [↗](#)). In *P. berghei*, the α 1-tubulin gene is expressed across most stages of the *Plasmodium* life cycle, whereas α 2-tubulin expression appears restricted to male gametocytes, gametes, and newly formed zygotes, indicating distinct functional roles for the two isoforms (32 [↗](#), 50 [↗](#)). Deletion of α 1-tubulin blocks sporozoite formation in oocysts and lowering α 1-tubulin expression leads to less infective sporozoites (50 [↗](#)).

There is a significant degree of amino acid conservation between human and *Plasmodium* tubulins, with approximately 83.7% identity for α -tubulin and 88.5% for β -tubulin. However, *Plasmodium* tubulin shows greater similarity to plant tubulin than to mammalian tubulin. This raises the potential for developing compounds that selectively target parasite microtubules while sparing the host cell cytoskeleton, offering a promising avenue for therapeutic intervention (43 [↗](#), 45 [↗](#)–47 [↗](#), 51 [↗](#)).

Recent studies on *Plasmodium falciparum* blood stage schizonts have revealed that hemi-spindles lack polyglutamylation, whereas subpellicular microtubules do exhibit this post-translational modification (52 [↗](#)–54 [↗](#)). Similarly, SPMTs appear to be stabilized by microtubule inner proteins, while spindle microtubules do not show these luminal proteins (55 [↗](#)). These differences may reflect functional specialisation of microtubule subpopulations in different cellular structures and developmental stages.

The liver stage of *Plasmodium* exhibits one of the fastest rates of replication known among eukaryotic cells, a process likely dependent on microtubule dynamics. However, to date, no studies have investigated microtubule dynamics during the liver stage. This study was therefore designed to explore the dynamics of parasite tubulin during the asexual liver stage, using recently developed expansion microscopy protocols (54,56–58).

Results

Expansion microscopy reveals the presence of the sporozoites subpellicular microtubules during the liver stage of infection

To investigate microtubule organization during the *Plasmodium* parasite liver stage, HeLa cells were infected with sporozoites and fixed them at different time points using cold methanol. We then performed IFA to stain host and parasite microtubules using an α -tubulin antibody (Figure 1A). The IFA revealed a marked accumulation of microtubules in proximity of the PVM (stained with anti-UIS4) from early infection (24 hours post-infection (hpi)) to the schizogony (30 to 54 hpi) (Figure 1A inlets). To examine these structures in greater detail, we applied the recently developed pre-staining expansion microscopy (PS-ExM) (56). By expanding the cells five-fold, PS-ExM enabled visualization of a large multimeric tubulin structure within the parasite, detectable from 6 hours post-infection (hpi) to 48 hpi (Figure 1B). Based on observations of sporozoites prior to liver infection, we hypothesize that this structure originates from the sporozoite subpellicular microtubules (SSPM) (Figure 1C top panel). We refer to this structure as the liver stage parasite microtubule bundles (LSPMB). Notably, the LSPMB progressively shrank over time and was no longer detectable by the end of schizogony (54 hpi) (Figure 1A, B). Using the 3D and 4D imaging software IMARIS, we quantified the LSPMB volume at various stages of liver infection. The results confirmed a consistent decrease in LSPMB size between 6 and 56 hpi (Figure 1D).

To verify that the LSPMB is not an artifact of the IFA and PS-ExM, we performed live cell imaging of infected HeLa cells. Cells were incubated with the tubulin tracker SPY650 for one hour prior to imaging. The LSPMB was clearly visible in sporozoites before invasion (Movie 1, ~10 hpi) and persisted until late liver stage development, gradually shrinking during schizogony (Figure 1E; Movies 2–4, ~36 hpi to 48 hpi). These live imaging results corroborate our findings from IFA and expansion microscopy, validating the presence and dynamic nature of the LSPMB during liver stage development.

The liver stage parasite subpellicular microtubules are dynamic

The SSPM has previously been described as highly stable structures, enabling sporozoite formation and invasion (50,55,59). These microtubules, similar to the corresponding microtubules in *T. gondii*, are resistant to conventional depolymerizing agents (59). To investigate whether the LSPMB retain this stability during liver stage development, we treated infected HeLa cells with the microtubule-depolymerizing agents Amiprophos-methyl (AMP) and oryzalin herbicide. These compounds are known to selectively target dynamic *Plasmodium* microtubules without affecting mammalian cells (43,45,47,51,60) (Figure 2 and Supplementary Figure 1A and B). We first validated drug specificity and cytotoxicity by testing various concentrations on HeLa cells over one week. Neither compound affected host cell viability (Supplementary Figure 1A and B) or disrupted host microtubules (Supplementary Figure 1D and E). However, oryzalin showed a stronger inhibitory effect on parasite growth compared to AMP (Supplementary Figure C) and was therefore selected for further experiments (Figure 2A). Sporozoites were dissected into medium containing 10 μ M or 20 μ M oryzalin and then used to infect HeLa cells (Figure 2A and B), which were incubated with the same drug concentrations. At 48 hpi, the cells were fixed and analyzed by IFA and PS-ExM. To test

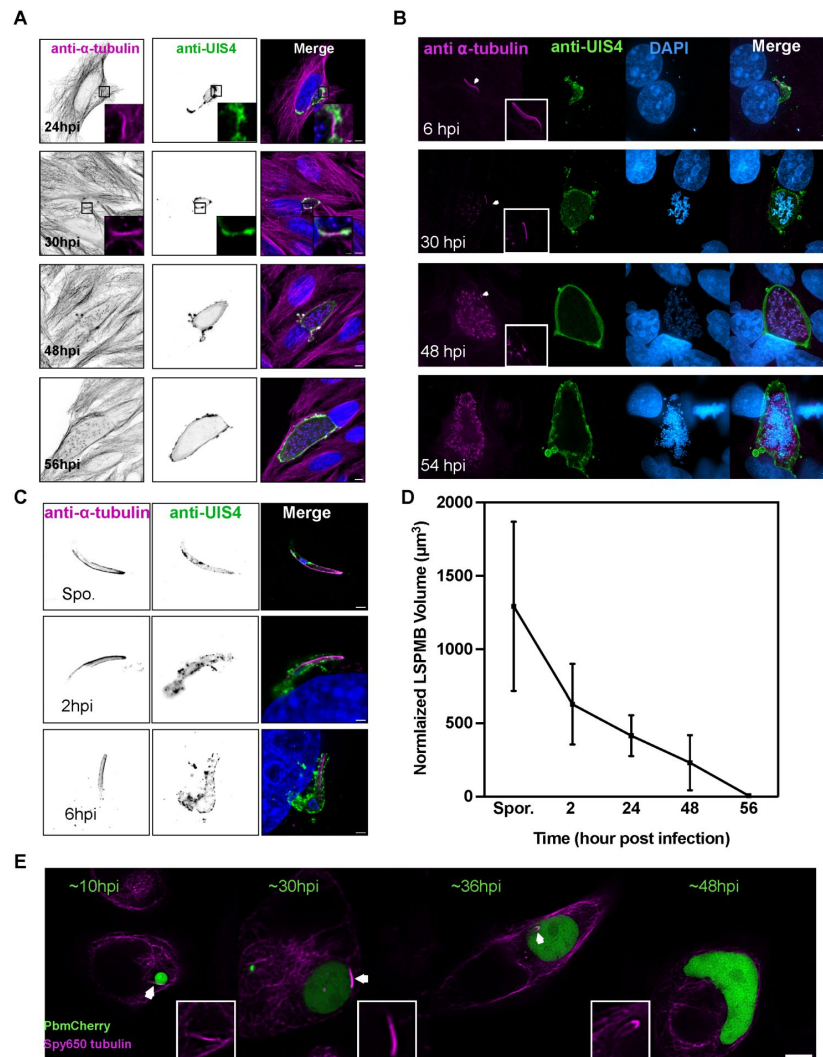


Figure 1

Immunofluorescence assay and expansion microscopy (PS-ExM) reveals liver stage parasite subpellicular microtubule.

(A), and (B), show maximum projection of z-stacks confocal images of infected HeLa cells and (C) sporozoites. Host and parasite microtubules were stained with anti- α -tubulin (magenta), and nuclei with DAPI (blue). The parasitophorous membrane vacuole (PVM) was stained with anti-UIS4 (green). (A) HeLa cells were infected with *Plasmodium* salivary glands sporozoites and methanol-fixed at 24, 30, 48, and 56 hpi. IFA was performed. (B) Fixed infected HeLa cells were expanded using the PS-ExM protocol. Inlets show LSPMB structures at 6, 30, and 48 hpi. Scale bar: 10 μm . (C) To determine whether the LSPMB originates from the sporozoite stage, PS-ExM was performed on salivary gland sporozoites, and parasite liver stage at 2 and 6 hpi. (D) LSPMB volume was quantified from PS-ExM z-stacks using 3D and 4D analysis in IMARIS software. Normalised mean volumes were: 1294.12 μm^3 in sporozoites, 628.34 μm^3 at 2 hpi, 414.84 μm^3 at 24 hpi, 231.4 μm^3 at 48 hpi, to 5.6 μm^3 at 54 hpi. (E) Live cell confocal imaging of infected HeLa cells incubated with tubulin tracker SPY650. Images show microtubules at 10, 30, 36, and 48 hpi. Host and parasite microtubules are shown in magenta; the parasite expresses cytosolic GFP (green). Scale bar: 5 μm .

reversibility, oryzalin was washed out at 24 hpi prior to the onset of schizogony (**Figure 2B**). At 10 μ M, oryzalin significantly impaired nuclear replication, spindle formation, and LSPMB volume when present the entire cultivation time (**Figure 2C, D** and **E**). Partial recovery of hemi-spindles and nuclear replication was observed, suggesting that the effect of 10 μ M oryzalin is reversible at this stage (**Figure 2C, D**, and **E**). However, at 20 μ M, neither the hemi-spindles nor the LSPMB recovered after drug removal, indicating irreversible microtubule disruption at higher concentrations (**Figure 2A** and **B**). To investigate whether the microtubule depolymerization by oryzalin affects the parasite growth and survival during the liver stage, we measured parasite size within infected HeLa cells at 6, 24 and 48 hpi. Treatment with 10 μ M oryzalin led to a significant reduction in parasite size and survival (**Figure 2 F, G, H**, and **I**). We further examined merozoite formation by staining for MSP1 at 56 hpi. Parasites treated with 10 μ M oryzalin failed to express MSP1, indicating a block in merozoite development and suggesting that intact microtubules are essential for DNA replication and merozoite formation (**Figure 2J**). Interestingly, at lower concentrations of 0.5 μ M, oryzalin treatment led to an unexpected elongation of the LSPMB and hemi-spindles up to four times longer than controls suggesting that microtubule turnover is tightly regulated (**Supplementary Figure 1F, G** and **H**). In conclusion, oryzalin-mediated depolymerization confirms that both the LSPMB and the hemi-spindles are dynamic structures during the liver stage. This dynamic behaviour was further supported by live-cell imaging using the SPY650 tubulin tracker (Movie 5, 6, and 7).

Given the observed microtubule remodeling in response to oryzalin, we next sought to determine whether known microtubule-associated proteins, specifically those regulating plus-end dynamics, are expressed during the liver stage and might contribute to the observed plasticity. Only a limited number of motor proteins and microtubule associated proteins, such as centrin and the kinesin protein family, have been identified during the *Plasmodium* liver stage (11–63). The recently characterized EB1 protein, known to regulate microtubules plus-end dynamics and length, has so far been reported for being expressed during the sexual blood stage (Mauer et al., 2023; Yang et al., 2023; Zeeshan et al., 2023). However, these studies did not assess EB1 expression during the asexual mosquito and liver stages. Given our observations of dynamic microtubule behavior and excessive polymerization following oryzalin treatment, we explored whether EB1 might be expressed during the liver stage. To this end, we generated a parasite line expressing endogenous EB1 fused to eGFP via double homologous recombination (**Supplementary Figure 2C** and **D**). As expected, EB1-eGFP was detected in sexual blood stages (**Supplementary Figure 2E** and **F**) consistent with previous reports (64,65). Interestingly, we also observed EB1 expression in both midgut and salivary gland sporozoites (**Supplementary Figure 2G**). Importantly, EB1 was also detected during liver stage development, where it localized to the LSPMB and hemi-spindles (**Supplementary Figure 2H–I**).

These findings suggest that EB1 may play a broader role in *Plasmodium* microtubule regulation than previously recognized, potentially contributing to the dynamic remodeling of microtubules during the liver stage.

The liver stage parasite subpellicular microtubule carries tubulin post-translational modifications

Microtubule functions are often regulated by different post-translational modifications (PTMs), which influence their stability, interactions, and dynamics. To gain insight into the dynamic nature of LSPMB during the liver stage development, we examined α -tubulin PTMs using immunofluorescence (IFA) and expansion microscopy (**Figure 3** and Movie 5, 6 and 7). To investigate whether the LSPMB is present in cell types other than HeLa cells, we infected primary hepatocytes with *Plasmodium* sporozoites and fixed the cells at different timepoints (6, 24, 36, 48, 54 and 56 hpi). We also analyzed sporozoites isolated from mosquito salivary glands. Using antibodies specific for polyglutamylated (anti-IN105) (66) (**Figure 3A**) and tyrosinated (anti-Ty) α -tubulin (**Figure 3B**), we observed that sporozoite microtubules, including the LSPMB, are

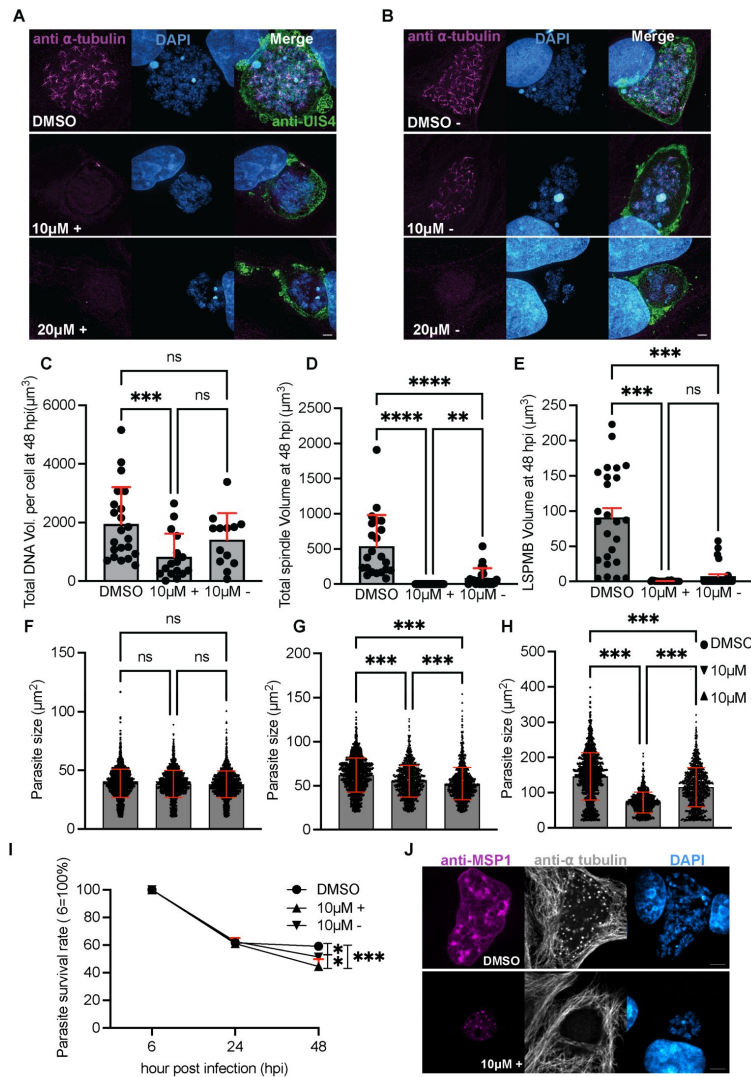


Figure 2

Microtubules are essential for *Plasmodium* liver stage development.

(A and B) show the effect of oryzalin on parasite microtubules during liver stage development. Sporozoites isolated from mosquito salivary glands were incubated in medium containing either 10 or 20 μM of oryzalin. HeLa cells were infected and fixed at 48 hpi with cold methanol prior to PS-ExM. (A) confocal z-stack projections of control cells (DMSO) and cells treated with 10 and 20 μM of Oryzalin. (B) confocal z-stack projections of infected cells where oryzalin was washed out at 24 hpi. Microtubule was stained with anti- α -tubulin (magenta), the PVM with anti UIS4 (green), and nuclei with DAPI (blue). Scale bar: 10 μm . “+” indicates cells treated with Oryzalin; “-” indicates cells from which the drug has been removed at 24 hpi. (C) Normalized total parasite DNA volume at 48 hpi. Mean DNA volume was 1952.02 μm^3 in the DMSO control, 827.26 μm^3 in 10 μM oryzalin-treated cells, and 1411.73 μm^3 after drug removal. (D) Normalized total hemi-spindle volume. The mean volume was 539.3 μm^3 in the control, 0 μm^3 with oryzalin treatment, and 89.08 μm^3 after drug removal. (E) Normalized LSPMB volume at 48 hpi. Mean volume was 5.92 μm^3 in the control, 2.3 μm^3 with oryzalin, and 1.65 μm^3 after drug removal. (F, G and H) Parasite size measurement at 6, 24, and 48 hpi, respectively. At 24 and 48 hpi, mean parasite size was 145.9 μm^2 in the control, 71.75 μm^2 with oryzalin, and 115.3 μm^2 after drug removal. (I) Parasite survival rate from 6 to 48 hpi. A slight increase in parasite death was observed in oryzalin-treated cells compared to the DMSO control. The red line on the graphs represents the mean and standard deviation. (C, D, E, F, and G), and (H): each dot represents an individual quantified parasite. (J) Confocal z-stack projection of IFA images at 56 hpi in cells treated with 10 μM oryzalin and DMSO control. Cells were fixed with cold methanol and stained for MSP1 (magenta), α -tubulin (grey) and nuclei (DAPI, blue). Scale bar: 5 μm . Statistical comparisons were performed using one-way ANOVA with Dunnett’s multiple comparisons test (* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$).

both polyglutamylated and tyrosinated. Quantitative colocalization analysis confirmed a strong overlap between LSPMB and polyglutamylated tubulin in both sporozoites and liver stage parasites (**Supplementary Figure 2A**). Interestingly, following hepatocyte invasion, the LSPMB lost tyrosination (i.e., became detyrosinated) while retaining polyglutamylation. This detyrosination persisted throughout liver stage development, including early and late schizogony (24–56 hpi). During this period, polyglutamylation was observed on the LSPMB and at the poles of the hemi-spindles, but not along the polymerizing hemi-spindles themselves, which eventually disappeared along with the LSPMB (**Figure 3A** and **C**). Given previous findings implicating polyglutamylation in hemi-spindle pole formation in erythrocytic merozoites (31), we investigated whether there is a structural relationship between the LSPMB and the hemi-spindle poles. Using PS-ExM and the tubulin tracker SPY650, we visualized direct connections between LSPMB filaments and hemi-spindle poles. Live cell imaging confirmed this connection between LSPMB and hemi-spindle poles, mainly during the shrinkage of the LSPMB (**Fig 3 F**; Movie 8, 9 and 10, Supplementary Movie 1). These observations suggest that the LSPMB may be redistributed to form or support the hemi-spindle poles during schizogony. We also stained for γ -tubulin and observed its localization at the hemi-spindle poles in both the parasite and host cell (**Supplementary Figure 2B**), supporting the idea that *Plasmodium* may utilize γ -tubulin for nuclear division and microtubule assembly, as reported in other systems (67–70).

To further define the spatial localization of the LSPMB, we used *P. berghei* parasites expressing the plasma membrane protein PMP1 fused to superfolder GFP (sfGFP) previously generated (71). ImageJ RGB profiling revealed that the LSPMB lies between the parasite plasma membrane (PM) and the parasitophorous vacuole membrane (PVM), suggesting that the sporozoite subpellicular microtubules undergo subcellular relocalization during liver stage development (**Figure 3D** and **E**). At later stages, particularly the cytome stage, we observed reappearance of tyrosinated tubulin near the parasite nuclei (**Figure 3B**). During merozoite formation, this tyrosinated tubulin appeared as punctate structures within the forming merozoites (**Figure 3B**).

Tyrosination has previously been associated with stable microtubules involved in motility, such as those in sporozoites, flagella, and spermatozoa (33,34,50,55). The absence of tyrosination during early intracellular liver stage development when the parasite is non-motile may facilitate its transformation from a banana-shaped sporozoite to a rounded liver stage form (**Figure 1** and **3**). LSPMB detyrosination may contribute to microtubule dynamic instability, providing both energy and tubulin monomers for rapid growth and spindle formation during schizogony. Finally, the reappearance of tyrosinated microtubules during merozoite formation may serve to stabilize the microtubules required for merozoite motility and erythrocyte invasion.

The *Plasmodium* α -tubulin C-terminal modification are important for sporozoite formation

To investigate the functional role of parasite α -tubulin PTMs during the liver stage development, we generated *P. berghei* mutant parasite lines with targeted mutations in the α -tubulin C-terminal region, where most PTMs are predicted to occur (**Figure 4A** and **Supplementary Figure 3A** and **B**). Specifically, we created three mutant lines: Ps Δ : a point mutation replacing the polyglutamylation site glutamic acid (E445) with alanine (A), Ty Δ : deletion of the tyrosination/detyrosination motif (YEADY) from position 449, Ct Δ : deletion of the entire C-terminal tail (EGEDEGYEADY) to prevent all possible C-terminal tail PTMs (**Supplementary Figure 3A** to **C**).

Using AlphaFold predictions, we confirmed that the mutations were unlikely to affect tubulin folding (**Supplementary Figure 3C**). The mutants were generated via double homologous recombination and verified by integration PCR and sequencing of the mutated α -tubulin gene (**Figure 4A** and **B**; **Supplementary Figure 3B**). The development of mutant parasites was investigated during the mosquito, and liver stages. Significant defects emerged during mosquito

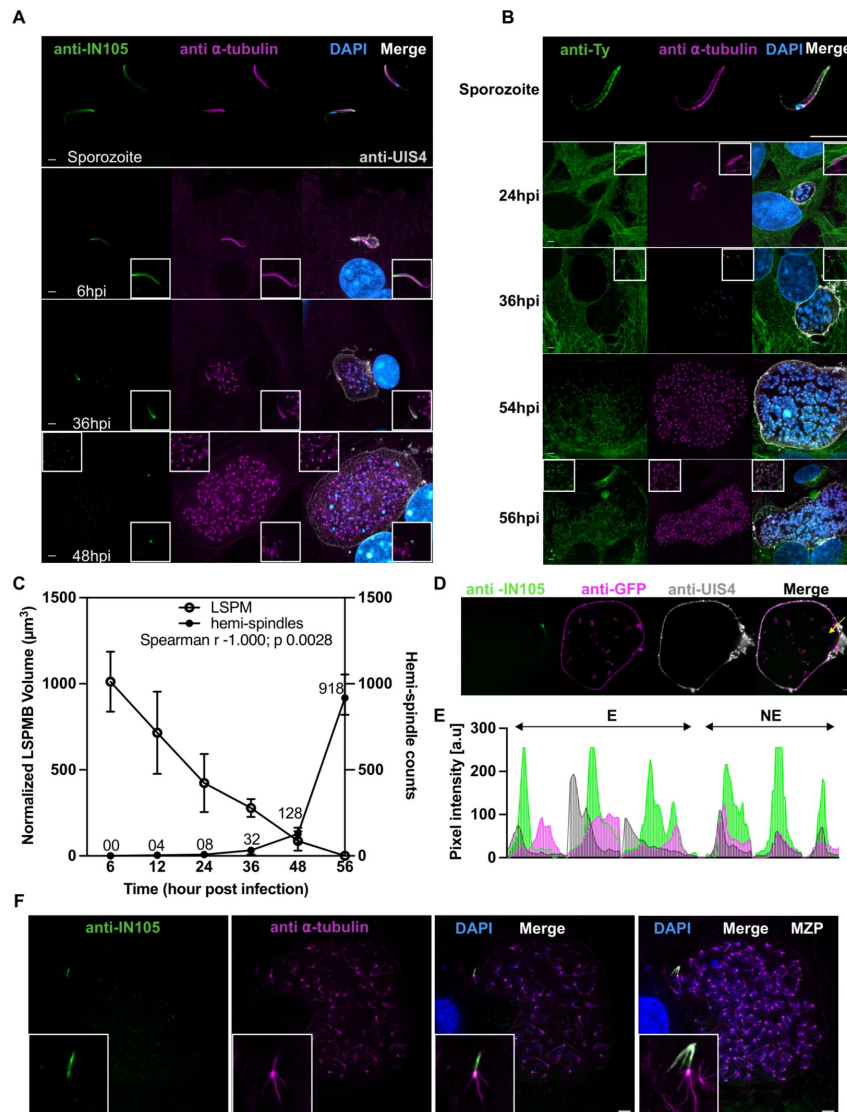


Figure 3

LSPMB is polyglutamylate.

(**A** and **B**) show maximum projection of confocal z-stacks from sporozoites and infected primary hepatocytes at different time points during liver stage development. Primary hepatocytes were isolated from mice and infected with sporozoites. Cells were fixed and stained with antibodies against (**A**) polyglutamylated tubulin (anti-IN105), (**B**) tyrosinated tubulin (anti-Ty), and α -tubulin (anti- α -tubulin). Nuclei were stained with DAPI (blue), and the PVM with anti-UIS4 (grey). ROIs in (**A**) show the LSPMB at 6 and 36 hpi and polyglutamylated hemi-spindle poles at 48 hpi. ROIs in (**B**) display LSPMB and α -tubulin evolution across liver stages. (**C**) Normalized LSPMB volume and the hemi-spindle pole counts from 6 hpi to 56 hpi, quantified using IMARIS 3D/4D analysis. LSPMB volume decreased from $1038.9 \mu\text{m}^3$ at 6 hpi to $0 \mu\text{m}^3$ at 56 hpi, while hemi-spindle poles count increased from 0 at 6 hpi to a median of 928 at 56 hpi, showing a perfect Spearman anti-correlation coefficient of -1. $n = 15$ per time point. (**D**) Single-plane PS-ExM confocal image of infected HeLa cells at 36 hpi with parasites expressing PMP1::GFP. Cells were PFA-fixed and stained with anti-GFP (magenta), anti-IN105 (green), and anti-UIS4 (PVM). The yellow arrow indicates the direction of Cell RGB Profiler analysis from outside the PVM to inside the parasite. (**E**) Pixel intensity profiles (arbitrary units) for PVM (grey), LSPMB (green), and parasite plasma membrane (magenta) in expanded (PS-ExM) cells (E) and non-expanded (NE) cells. Overlap was observed in NE cells, while PS-ExM resolved LSPMB localization between the PVM and the parasite plasma membrane. $n=10$, in triplicate. Scale bar: $10 \mu\text{m}$. (**F**) Single-plane confocal image of infected primary hepatocytes fixed with cold methanol. Cells were stained with anti-IN105 (green), α -tubulin (magenta), and DAPI (blue). The ROI shows a LSPMB tubule in direct contact with a parasite hemi-spindle pole. MZP: Maximum Z projection. Scale bars: $10 \mu\text{m}$.

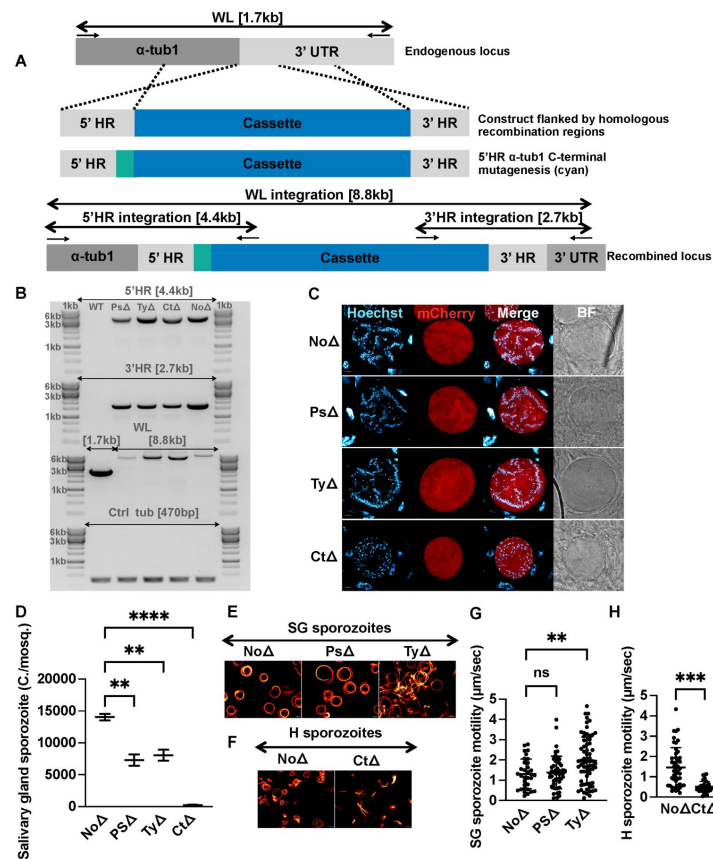


Figure 4

***Plasmodium* α 1-tubulin C-terminal post-translational modification is important for the mosquito stage development.**

(A) Schematic of the double homologous recombination (DHR) construct targeting the *Plasmodium* α 1-tubulin gene (PBANKA_0417700) for C-terminal mutagenesis. The cassette (blue box) includes the 3'UTR *P. berghei* dihydrofolate reductase (PbDHFR), elongation factor-1 alpha (ef1 α) promoter, human dihydrofolate reductase (hdhfr), yeast FCU (yfcu), 3'UTR PbDHFR, Hsp70 promoter, mCherry and the 3' UTR Hsp70, with the 3'UTR region of the PBANKA_0417700 gene. This cassette was flanked by a 5' homologous region (5'HR) containing the α 1-tubulin coding region including the stop codon (with introduced mutations in the C-terminal region, cyan box) and a part of the 3'UTR of the gene (3'HR). The construct was transfected into the PbWT (*P. berghei* ANKA line cl15cy1). Black arrows indicate PCR primer binding sites and expected product sizes. Primer sequences are listed in **Table 2** and 3. **(B)** Integration PCR results for the mutant parasite lines, showing expected band sizes on a 1% agarose gel: 5'HR = 4.4 kb, 3'HR = 2.7 kb, whole locus (WL) = 8.8 kb. Ladder: 1 kb DNA marker. Genotypes: WT, (wild-type); Ps Δ (glutamine-to-alanine substitution); Ty Δ (deletion the C-terminal tyrosination/detyrosination residues); Ct Δ (complete C-terminal deletion), and No Δ (construct control with no mutation). To ensure that all samples contained the same amount of DNA, a control gene representing a portion of tubulin (Ctrl tub about 470 bp) was used for integration PCR. **(C)** Representative confocal images of live oocysts at 11 days post infection (dpi). Nuclei were stained with Hoechst 33342 (hot cyan) and cytosolic mCherry (red) was expressed under the HSp70 promoter. BF, brightfield. **(D)** Normalized salivary sporozoite counts at 18 dpi. The control (No Δ) averaged 14,033 sporozoites count per mosquito (C./mosq.), compared to 7,300 (Ps Δ), 8,066 (Ty Δ), and 200 (Ct Δ). Experiment performed in triplicate; n = 20 per replicate. **(E and F)** Wide-field microscope motility assay images of salivary gland sporozoites showing the typical circular movement (maximum z projection in Fiji) of the control, Ps Δ , and Ty Δ salivary gland (SG) sporozoites **(E)** and the control and Ct Δ hemolymph (H) sporozoites **(F)**. For Ct Δ , hemolymph sporozoites were used due to the absence of salivary gland sporozoites; these lacked typical motility. **(G)** Quantification of sporozoite gliding velocity for Ps Δ , and Ty Δ salivary gland sporozoites. Mean velocities: No Δ = 1.3 μ m/s, Ps Δ = 1.38 μ m/s, Ty Δ = 1.9 μ m/s, Ct Δ = 0.5 μ m/s. Statistical analysis: one-way ANOVA with Dunnett's multiple comparisons test (*p $\leq$ 0.05, **p $\leq$ 0.01, ***p $\leq$ 0.001). **(H)** Quantification of sporozoite gliding velocity of Ct Δ haemolymph sporozoites. The average gliding speed for control hemolymph sporozoites was 1.4 μ m/s, whereas Ct Δ sporozoites exhibited a slower speed of 0.5 μ m/s. Statistical comparison tests were performed using two tailed unpaired t tests (*p $\leq$ 0.05, **p $\leq$ 0.01, ***p $\leq$ 0.001).

stage development. The PsΔ and TyΔ mutants were able to form oocysts (**Figure 4 C** [↗](#), **Supplementary Fig 4A** [↗](#)) and gliding sporozoites, albeit in reduced numbers (**Figure 4D**, **E** [↗](#) and **G** [↗](#); **Supplementary Fig 4B** [↗](#); and Supplementary Movie 2, 3, 4 and 5). Quantification of salivary gland sporozoites at 18 days post-blood feeding showed that PsΔ and TyΔ mutants produced approximately half as many sporozoites as the control line (NoΔ) (**Figure 4D** [↗](#)). We next examined sporozoite motility, which typically consists of continuous circular gliding (16 [↗](#), 72 [↗](#)–76 [↗](#)). PsΔ and TyΔ salivary gland sporozoites were assessed and compared to controls. While PsΔ have a similar gliding velocity compared to controls, TyΔ sporozoites displayed increased gliding velocity (2.5 μm/s vs. 1.4 μm/s) (**Figure 4G** [↗](#)), suggesting a regulatory role of the tyrosination motif in fine-tuning motility. To assess whether these motility phenotypes influenced infectivity, we performed in vivo infection assays with PsΔ and TyΔ. All mutant lines sporozoites were able to infect mice similarly to control parasites (**Supplementary Figure 4C** [↗](#)). However, the CtΔ mutant failed to produce salivary gland sporozoites entirely (**Figure 4D** [↗](#)). Since CtΔ mutant parasites failed to reach the salivary glands, we analyzed hemolymph sporozoites for this line compared to the controls (**Figure 4F** [↗](#)). These showed severely impaired motility, lacking the typical gliding pattern and exhibiting significantly reduced speed (**Figure 4F** [↗](#) and **H** [↗](#)). Given that PsΔ and TyΔ mutants produced motile salivary gland sporozoites, we infected HeLa cells with PsΔ and TyΔ mutant sporozoites and CtΔ mutant hemolymph sporozoites to assess their liver stage development and identify any downstream phenotypic effects.

The *Plasmodium* α-tubulin C-terminal mutants undergo liver stage development

Using polyglutamylation-specific antibodies, we found that PsΔ mutant parasites still exhibited polyglutamylated liver stage parasite microtubule bundles (LSPMBs) in both sporozoites and liver stage forms, although the signal intensity was significantly reduced compared to controls (**Figure 5A**, **B** [↗](#) and **C** [↗](#) and **G** [↗](#)). As tyrosination was previously observed only in sporozoites and late liver stages, we performed PS-ExM on sporozoites and stained them with α-tubulin tyrosination-specific antibodies. TyΔ salivary gland sporozoites and CtΔ hemolymph sporozoites both showed detectable tyrosination signals, although the signal in CtΔ parasites was markedly weaker than in controls (**Figure 5D**, **E** [↗](#) and **F** [↗](#)). Interestingly, as PTM-specific antibodies continued to recognize signals in the PsΔ and TyΔ mutants, it is possible that the parasite compensates by relocating the PTM sites to adjacent residues within the C-terminal sequence. Alternatively, the antibody signal may stem from β-tubulin modifications rather than α-tubulin.

To assess parasite survival and growth during liver stage development, we performed quantitative microscopy (**Figure 5H** [↗](#) to **O** [↗](#)). Both PsΔ and TyΔ mutant parasites successfully infected HeLa cells and progressed through liver stage development (**Figure 5B**, **D**, **H**, **I**, **J** [↗](#) and **K** [↗](#)). This was further confirmed by staining the parasitophorous vacuole membrane (PVM) using anti-UIS4 antibodies (**Figure 5A**, **B** [↗](#) and **C** [↗](#)). Parasite size was measured at 6 hpi and 48 hpi to assess developmental progression. No significant differences were observed at 6 hpi (**Figure 5I** [↗](#)). However, by 48 hpi, mutant parasites were significantly smaller than controls, indicating impaired growth (**Figure 5J** [↗](#)). Survival rates of PsΔ and TyΔ mutants were comparable to controls (**Figure 5K** [↗](#)), with infection rates of 15% for PsΔ (matching the control) and 27% for TyΔ, exceeding the control. As expected, the CtΔ hemolymph sporozoites failed to infect and to develop liver stage forms (**Figure 5C**, **L**, **M**, **N** [↗](#) and **O** [↗](#)). The sporozoites failed to establish productive infections with a typical PVM at all examined time points (**Figure 5C**, **M**, **N** [↗](#) and **O** [↗](#)).

To confirm the phenotypes, we compared our mutants with previously generated *P. berghei* α1-tubulin mutants described by Spreng et al. (2019), including parasites with intron deletions (intronΔ), C-terminal tyrosination deletions 540ADY (C-termΔ), and α2-tubulin exon swaps. These mutants showed similar phenotypes, including a complete absence of salivary gland sporozoites in

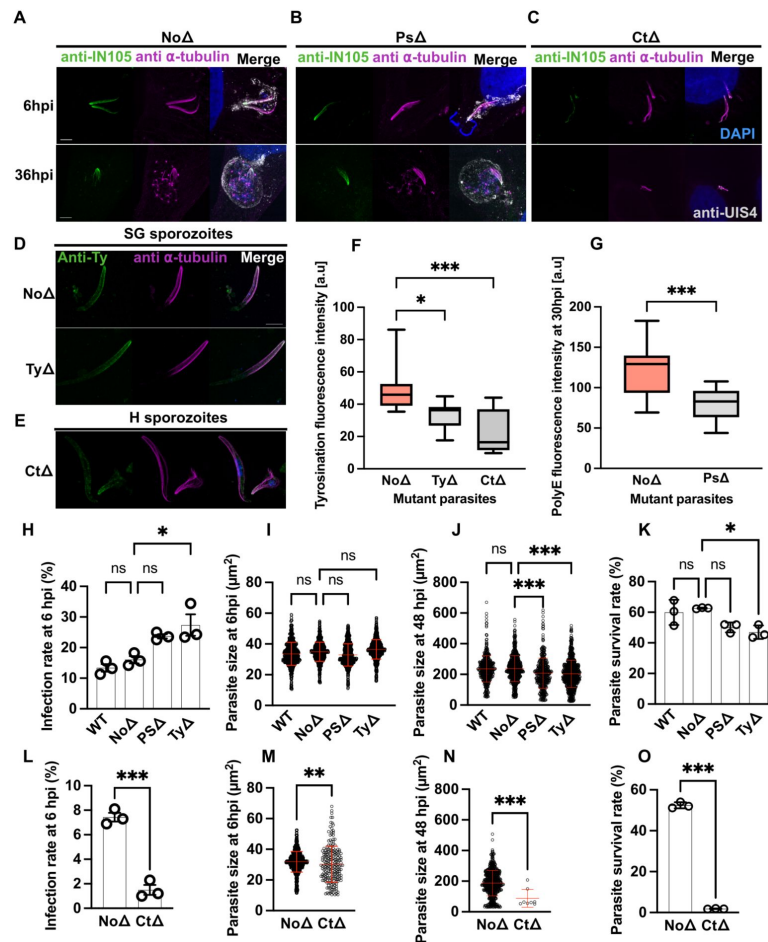


Figure 5

Liver stage development of *Plasmodium* α1-tubulin mutant parasites.

(**A**, **B** and **C**) Maximum projections of PS-ExM-acquired z-stack confocal images of infected HeLa cells at 6 and 36 hpi, showing the control (NoΔ), PsΔ, and CtΔ mutant parasites. (**D** and **E**) Maximum projections of PS-ExM-acquired z-stack confocal images of control and TyΔ salivary gland sporozoites, as well as CtΔ hemolymph sporozoites (**E**). In (**A**, **B** and **C**), cells were stained with antibodies against polyglutamylated tubulin (anti-IN105), while (**D** and **E**) antibodies against tyrosinated tubulin (anti-Ty) were used. In all panels, α-tubulin was labeled with anti-α-tubulin, nuclei with DAPI (blue), and the parasitophorous vacuole membrane (PVM) with anti-UIS4 (grey). (**F** and **G**) represent the fluorescence intensity (arbitrary units, a.u.) of anti-Ty in sporozoites (**F**) and anti-IN105 at 36 hpi in the liver stage (**G**). In (**F**), the mean fluorescence intensity of CtΔ (22 a.u.) was significantly reduced compared to TyΔ (33.4 a.u.) and NoΔ (49.6 a.u.), while in (**G**), PsΔ (79.9 a.u.) showed a significant reduction compared to NoΔ (121 a.u.). (**H** and **L**) showed infection rates of mutant parasites in HeLa cells at 6 hpi, determined by pre-counted sporozoites and quantitative microscopy. PsΔ (23.7%) and TyΔ (27.3%) exhibited typical infection rates compared to control (NoΔ) (16%) and WT (14%) (**H**), whereas CtΔ hemolymph sporozoites had a drastically reduced infection rate (2%) relative to controls hemolymph sporozoites (7.4%) (**L**). (**K** and **O**) Parasite survival from 6 to 48 hpi, normalized to 100% at 6 hpi. At 48 hpi, survival rates were 63% (NoΔ), 60% (WT), 50% (PsΔ), and 47% (TyΔ) (**K**). In contrast, CtΔ hemolymph sporozoites showed a significantly lower survival rate (2%) compared to controls (52%) (**O**), indicating impaired viability. (**I** and **J**, **M** and **N**) Parasite size measurements at 6 and 48 hpi for control, PsΔ, and TyΔ salivary gland sporozoites (**I** and **J**), as well as control and CtΔ hemolymph mutants (**M** and **N**). At 48 hpi, PsΔ (208.16 μm²) and TyΔ (203.7 μm²) were significantly smaller than NoΔ (237.9 μm²) and WT (235.7 μm²) (**J**). The CtΔ hemolymph mutants failed to establish liver stage infection, with an average size (88.1 μm²) markedly smaller than controls (187.4 μm²) (**N**). Each dot represents an individual parasite. PsΔ: Glutamine-to-alanine substitution at the polyglutamylation site; TyΔ: Deletion of C-terminal tyrosination/detyrosination residues; CtΔ: Complete deletion of PBANKA_0417700 α-tubulin C-terminal PTM residues and NoΔ: Control construct with no mutation. WT: wild type parasite. The experiments were conducted using, (**F**, **H**, **I**, **J**, **K**): One-way ANOVA with Dunnett's multiple comparisons test (*p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001). (**G**, **L**, **M**, **N**, **O**): Two-tailed unpaired t-tests (*p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001). All experiments were performed in triplicate.

α 2-tubulin exon swap parasites, mirroring the Ct Δ mutant phenotype (**Figure 4D** and **H**; **Supplementary Figure 3D** and **E**). These results underscore the importance of α -tubulin C-terminal PTMs in sporozoite formation and migration from the midgut to the salivary glands.

The failure of Ct Δ parasites to generate viable sporozoites highlights the essential role of α -tubulin PTMs in sporozoite development and migration. Co-mutation of both α - and β -tubulin PTM sites may prove lethal to parasite development in the mosquito stage. Future studies should focus on identifying and characterizing the enzymes responsible for tubulin PTMs in *Plasmodium*, and genetic knockout strategies should be pursued to further elucidate the function of these modifications.

Discussion

Our study provides the first in-depth characterization of microtubule architecture and dynamics during the *Plasmodium* liver stage. Using tubulin trackers and α -tubulin antibodies, we identified two major microtubule populations: a large, multimeric subpellicular network that we termed the liver stage parasite microtubule bundles (LSPMB), and the polymerizing hemi-spindles involved in nuclear division. The LSPMB appears to originate from the SSPM and persists until late schizogony. These observations align with earlier electron microscopy studies by Meis et al. (77) who reported the presence of subpellicular microtubules and inner membranes in liver stage parasites up to the point of mitotic spindle formation.

The behavior and function of microtubules are closely regulated by PTMs. We found that α 1-tubulin expressed during the liver stage carries residues associated with polyglutamylation and tyrosination. Using PTM-specific antibodies, we confirmed that *Plasmodium* sporozoites display both modifications on their subpellicular microtubules. Tyrosination, a well-known marker of stable microtubules, has been described in highly motile systems such as cilia, flagella, and spermatozoa (33,34,50,55). This is consistent with the need for high motility in sporozoites, which must traverse multiple tissue barriers during transmission (16,72,73).

Polyglutamylation, in contrast, introduces negative charges on the tubulin surface that enhance interactions with MAPs, and is commonly associated with dynamic microtubule behavior (78–80). This modification has previously been linked to merozoite formation and intracellular trafficking in *Plasmodium* blood stages (31). The co-occurrence of tyrosination and polyglutamylation in sporozoites likely ensures a microtubule network that is both stable and functionally interactive during mosquito stage migration.

Upon hepatocyte invasion, we observed a loss of tyrosination while polyglutamylation was maintained, suggesting a functional switch from stable to dynamic microtubules (**Figure 6**). This transition may facilitate morphological transformation from a motile, slender sporozoite into a rounded liver stage form and could also support the parasite's rapid replication. The liver stage is one of the fastest-replicating phases in the *Plasmodium* life cycle, with replication rates among the highest observed in eukaryotic cells (81,82). During this phase, the parasite synthesizes proteins de novo and hijacks host cell lipids and proteins for membrane biogenesis (83–86). We propose that the parasite brings its SSPM into hepatocytes to be repurposed as the LSPMB (**Figure 6**). The observed microtubule disassembly over time may support tubulin turnover and save energy for intracellular development.

In addition to their structural roles, LSPMB filaments were occasionally found in contact with hemi-spindle poles, suggesting that LSPMB contribute to spindle formation during schizogony. This connection, although infrequent in fixed cells, was frequently observed during live imaging suggesting a transient connection. We also detected expression of EB1, a microtubule plus-end

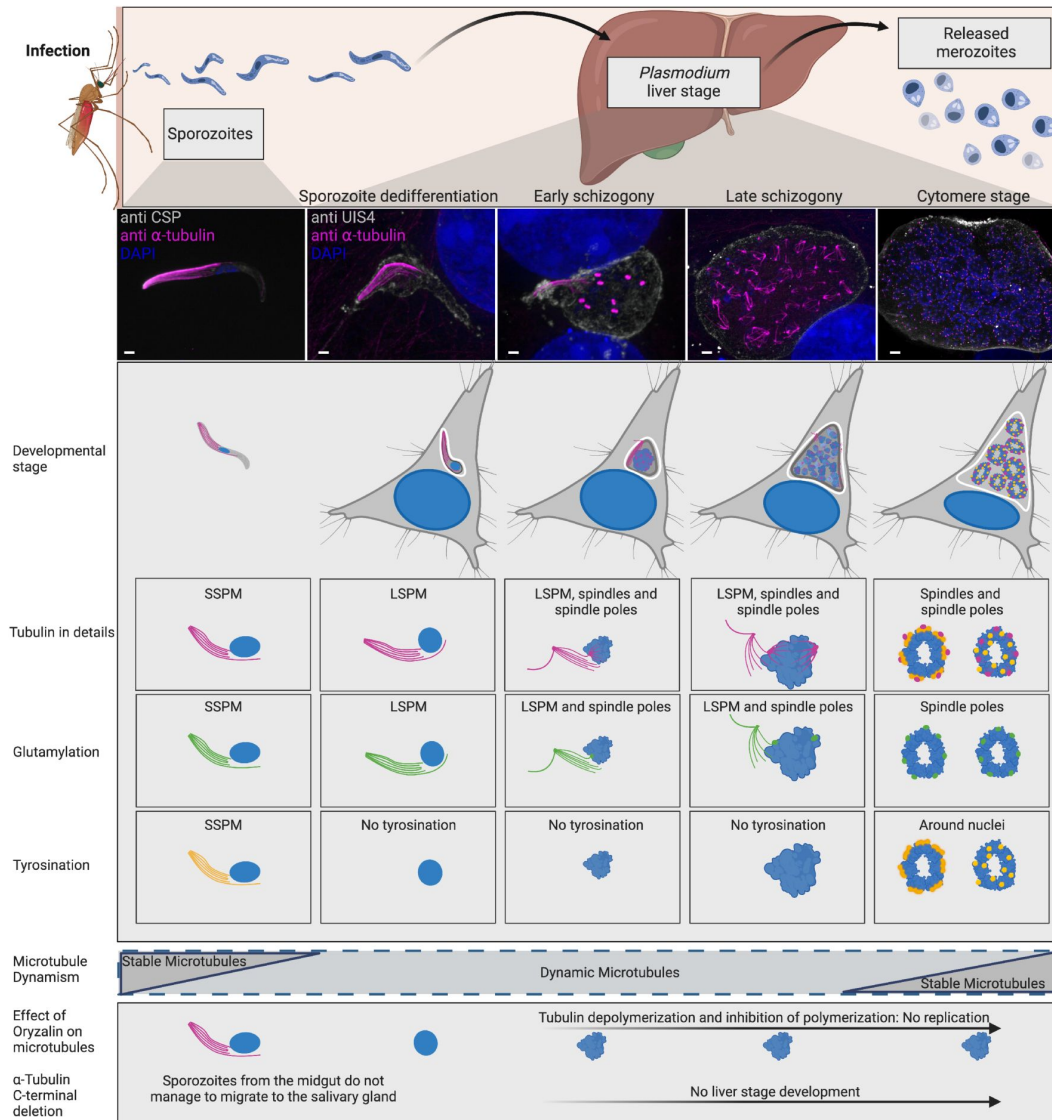


Figure 6

Model of the tubulin behavior during *Plasmodium berghei* salivary gland sporozoite and exo-erythrocytic stage.

The graphical abstract summarizes the key finding of this study. It illustrates the development of *Plasmodium* parasites within primary hepatocytes, from invasion to the cytomere stage. α -tubulin was stained with anti- α -tubulin (magenta), nuclei with DAPI (blue), and the parasitophorous vacuole membrane (PVM) with anti-UIS4 (grey). The model highlights the evolution of sporozoite subpellicular microtubules (SSPM) into liver stage subpellicular microtubules (LSPMB), along with their associated tubulin post-translational modifications (PTMs). We propose that polyglutamylated LSPMB originates from the SSPM and is recycled during liver stage development to support hemi-spindle pole formation, membrane biogenesis, and potentially other functions. Upon hepatocyte invasion, the SSPM loses tyrosination, a modification that initially confers stability and supports sporozoite motility. This detyrosination likely facilitates parasite shape transformation, increases microtubule dynamics, and enables interactions with ER structures and MAPs. Tyrosination reappears in late stages, possibly stabilizing merozoite microtubules for erythrocyte invasion. Microtubule-stabilizing drugs did not affect the SSPM but inhibited liver stage microtubule polymerization, confirming their dynamic nature. Inhibition of microtubule dynamics also prevented merozoite formation. Finally, C-terminal tubulin PTM residues were shown to be essential for salivary gland sporozoite development and successful liver stage infection.

binding protein previously believed to be restricted to the sexual blood stage (64,65,87), during mosquito and liver stages. Its localization to LSPMB and hemi-spindles suggests a broader role for EB1 in regulating microtubule dynamics throughout the parasite life cycle.

To directly test the function of tubulin PTMs, we generated *P. berghei* mutants with targeted C-terminal modifications of α 1-tubulin. These included a glutamate-to-alanine substitution at a polyglutamylation site, deletion of the tyrosination motif, and complete deletion of the C-terminal tail. Despite the mutations, PTM-specific antibody staining remained detectable in the first two mutants, suggesting compensatory mechanisms such as modification of adjacent residues or of β -tubulin. The tyrosination antibody used (anti-Ty, T9028, Sigma-Aldrich) is a purified monoclonal antibody targeting C-terminal tyrosine residues on tubulins, whereas the monoclonal GT335 antibody (AG-20B-0020-C100, AdipoGen Life Sciences) recognizes the epitope formed at the branching point of polyglutamylation side chains (GEGE*EE, * marking the modification site) and detects both short and long glutamate side chains. The polyclonal antibody (AG-25B-0030-C050, AdipoGen Life Sciences) was raised against at least three glutamate residues and is thereby specific to long polyglutamylation side chains, though both antibodies may cross-react with β -tubulin or other proteins containing genetically encoded C-terminal modifications (66,88). The reduced fluorescence signal observed (**Figure 5**) in mutant parasite microtubules suggests either antibody cross-reactivity with β -tubulin or compensatory PTM relocation to adjacent α -tubulin residues. In the Ct Δ mutant (lacking the entire C-terminal tail), PTM relocation is abolished, and residual signal likely stems from β -tubulin which was insufficient to sustain sporozoite motility and transmission. These possibilities warrant further investigation. Previous studies have also demonstrated partial redundancy between α 1- and α 2-tubulin modified (49,50,89). However, the complete C-terminal deletion mutant failed to produce salivary gland sporozoites and could not establish liver stage infection. These results are consistent with findings from Spreng et al. (2019), who showed that α 1-tubulin mutants lacking the C-terminal region produced defective sporozoites with altered microtubule numbers and impaired motility.

We also observed that the LSPMB relocates during liver stage development. Using expansion microscopy on PMP1::GFP-expressing parasites (71), we found that the LSPMB localizes between the parasite plasma membrane and the PVM. This differs from earlier reports of SSPM positioning (77,90–92) but both notions are not mutually exclusive. It is possible that the LSPMB first localizes within the parasite but then re-localizes at later stages or even re-localizes dynamically. Its localization between the PM and the PVM positions it well for interaction with the host cell indicating an interaction with host cell structures such as ER tubules or host microtubules (71,84,93,94). However, whether this re-localization plays a role in nutrient acquisition or host-pathogen signaling remains to be explored.

Finally, during late schizogony and merozoite formation, we observed the reappearance of tyrosinated tubulin. This may reflect the need for stable microtubules in newly formed merozoites, which must be competent for host cell invasion.

In conclusion, our study reveals a previously unrecognized complexity in the microtubule biology of *Plasmodium* liver stages. The identification of the LSPMB as a dynamic structure regulated by tubulin PTMs and repurposed from the sporozoite cytoskeleton opens new avenues for understanding how the parasite controls its intracellular architecture. These PTMs appear critical for parasite shape, replication, and stage transition, and may represent valuable targets for therapeutic intervention. Future work should aim to identify the enzymes responsible for these modifications and explore their potential as drug targets. The behavior and function of microtubules are closely tied to their PTMs. We found that α 1-tubulin expressed during the liver stage carries residues predicted to undergo polyglutamylation and tyrosination.

Using PTM-specific antibodies, we confirmed that sporozoite microtubules are both tyrosinated and polyglutamylated. Tyrosination is typically associated with stable microtubules and has been described in highly motile structures such as cilia and spermatozoa. This is consistent with the sporozoite's need for high motility gliding at approximately 1.5 $\mu\text{m/s}$ as it migrates from the mosquito midgut to the salivary glands and eventually traverses skin and vasculature to reach hepatocytes.

Methods

Ethics statement

All experimental procedures were performed in full compliance with the Swiss Animal Protection Act (TSchG) and were approved by the Canton of Bern's Animal Experimentation Commission (Authorization, BE118/22). Personnel involved in animal studies held the Federation of European Laboratory Animal Science Associations (FELASA) accredited certifications. Female C57BL/6 and BALB/c mice of 6 to 10 weeks old and 20–30 g at infection were sourced from Janvier Labs (France) or bred in the animal facility at the University of Bern's Institute of Cell Biology. Animals were maintained under standard housing conditions with daily health monitoring by trained staff.

P. berghei lines used in this study and the generation of transgenic parasites

The *P. berghei* ANKA parasite line was utilized for both murine and cellular infection models. To generate the $\alpha 1$ -tubulin C-terminal mutant and eb1::eGFP parasite lines, the C-terminal coding and 3' untranslated regions (UTRs) were cloned into the double homologous recombination vector pDHR, which contains the 3'UTR PbDHFR, the ef1 α promoter region, the selectable markers hdhfr::yfcu, and additional 3'UTR PbDHFR sequences. The pDHR vector backbone originated from the plasmid pG362 (kindly provided by the Waters lab) (95 [DOI](#)).

First, the pG362 plasmid was digested with SacII and PstI to remove the AID, selectable marker, and GFP sequences. The amplified product (linker-eGFP-3'UTR PbDHFR) from the pOB277 vector was then cloned using T4 DNA ligase (NEB, M0202L). For subsequent cloning, restriction sites were added to the forward (KpnI and ApaI) and reverse (EcoRI and SmaI) primers. The plasmid was then digested with EcoRI and SmaI to clone the amplified product: ef1 α promoter region-hdhfr::yfcu marker (a fusion gene of the positive selection marker human dihydrofolate reductase (hdhfr) and the negative selection marker yeast cytosine deaminase-uridyl phosphoribosyl transferase (yfcu) 3'UTR PbDHFR from the genomic DNA of the *P. berghei* GIMO parasites (RMgm-687) parental line (96 [DOI](#)).

The complete $\alpha 1$ -tubulin gene sequence (PBANKA_0417700), including 5' and 3' UTRs, was retrieved from PlasmoDB (<https://plasmodb.org> [DOI](#)). The linker-eGFP region of the pDHR vector was removed using ApaI and NotI restriction enzymes. The 5'UTR of the $\alpha 1$ -tubulin gene was then subcloned into the digested pDHR using Gibson assembly. The resulting vector was digested with SmaI, and the PbHsp70 promoter and mCherry regions from RMgm-687, along with the 3'UTR of the $\alpha 1$ -tubulin gene, were subcloned into the vector at the ApaI and SmaI restriction sites, respectively, using Gibson assembly.

Site-directed mutagenesis (QuickChange II XL Kit) was used to introduce the C-terminal modifications, with primer sequences listed in **Tables 2** [DOI](#) and **3** [DOI](#). The final plasmid was linearized with ScaI before being transfected into the PbWT line (*P. berghei* ANKA line cl15cy1) (97 [DOI](#)). Similarly, the end-binding protein eb1 (PBANKA_0405600) with an eGFP tag was engineered using identical cloning strategies. The 5' and 3' UTRs were retrieved from PlasmoDB (<https://plasmodb.org> [DOI](#)) and subcloned into the pDHR vector at the ApaI and SmaI restriction sites,

respectively, using Gibson assembly. The final vector was linearized with NdeI before being transfected into the Pb1868 line (constitutively expressing Hsp70-mCherry-ef1α-NLuc) (98). Primer sequences are listed in Table 1.

Vector transfections were performed in schizont cultures as previously described (16,99). Briefly, *P. berghei* schizonts were cultured and purified (16,99,100). Electroporation of schizonts from the respective parasite lines was performed using the Amaxa® Human T Cell Nucleofector® system with the U33 program, followed by intravenous injection into naïve mice. The construct was integrated by double-crossover homologous recombination at the locus of interest. Pyrimethamine drug selection was administered via drinking water, and parasitemia was monitored daily from day 7 post-infection until mice tested positive (99).

Parasitaemia measurement and cloning of transgenic parasite lines

To measure the parasitemia ratio of infected mice, approximately 2 µL of blood from the tail vein of each infected mouse was added to 500 µL of PBS and analyzed by flow cytometry using the NovoCyte (Agilent, 2023) as described (101). The data were analyzed using NovoExpress Flow Cytometry Analysis Software (version 1.6.2, Build 230717.50551; Agilent Technologies).

To clone the different engineered transgenic parasite lines, mice were infected with the transgenic parasites. Three days post-infection, parasitemia was assessed and when parasitemia reached at least 1%, blood was collected from the infected mouse via heart puncture following euthanasia as described (16). The collected blood was passaged into a new naïve mouse. When parasitemia in the newly infected mouse reached 1-2%, 2 µL of tail vein blood was collected and mixed with RPMI-1640. Maintaining low parasitemia was crucial to prevent both double erythrocyte infection by *P. berghei* and gametocyte formation. Single *P. berghei*-infected erythrocytes were then sorted into individual wells of 96-well plates (Cellvis) containing 200 µL of RPMI-1640 supplemented with 10% FCS using a BD FACSDiscover S8 with CellView. Supplementing the medium with 10% FCS allowed the sorted infected erythrocytes to recover from the sorting process. Five mice were then intravenously infected with a single sorted infected erythrocyte each. The mice underwent drug selection via drinking water, and parasitemia was monitored daily from day 7 post-infection until parasites were detected. Clonality of the parasites in positive mice was verified by integration PCR using primers listed in Tables 1, 2, and 3.

Cell culture

Human epithelial HeLa cells (European Cell Culture Collection) were maintained in Minimum Essential Medium (MEM; 1-31F01-I BioConcept) supplemented with Earle's salts and 10% heat-inactivated fetal bovine serum (FBS; Sigma-Aldrich), hereafter referred to as complete MEM (cMEM). Primary hepatocytes were cultured in William's Medium E (WME) complete medium. All cells were incubated at 37°C under 5% CO₂. For infection assays, 40,000 HeLa cells were seeded onto coverslips in 24-well plates and allowed to adhere overnight. Cells were then infected with *P. berghei* sporozoites and maintained in cMEM containing 2.5 µg/mL amphotericin B (PAA Laboratories, P11-001) to prevent fungal contamination. Medium was exchanged every 24 hours.

Primary hepatocyte isolation

The mice primary hepatocytes were isolated according to the method described (8). Briefly, the cells were isolated from 12 to 16 week old Balb/c or C57BL/6 mice using a two-step perfusion technique (8,83,102). Mice were euthanized via CO₂ inhalation (1–2 minutes), and after securing the animal in a supine position, a midline abdominal incision was made to expose the portal vein or inferior vena cava (IVC). A peristaltic pump was pre-rinsed with 70% ethanol and HEPES buffer, then set to 50 for initial cannulation of the portal vein (or IVC) using a Teflon-coated needle. Upon successful insertion, the flow was increased to maximum (999), and the opposing

Table 1

***Plasmodium berghei* eB1egfp parasites integration PCR primer**

Integration	Mutants	Primer pair	Primer Sequence	Expected size
5'HR		upstream of the construct	GGTTGTAAGTCGAAAAGAGCT GATAGAA	1.5kb
		within the ef1alpha	TCGATGCCCTTCAGCTCG	
3'HR		within yfcu	GAGGTGTTAAGCCAGAGAG	1.3kb
		downstream the 3'UTR	CGGTTCAATAAATGAGGGATTA GC	
Whole locus		Forward	GGTTGTAAGTCGAAAAGAGCT GATAGAA	6kb
		Reverse	CGGTTCAATAAATGAGGGATTA GC	
Control		Forward	TGGAGCAGGAAATAACTGGG	470bp
		Reverse	ACCTGACATAGCGGCTGAAA	

Table 2

***Plasmodium berghei* α1-tubulin mutant parasites integration PCR primer**

Integration	Mutants	Primer pair	Primer Sequence	Expected size
5'HR	PsΔ TyΔ CtΔ NoΔ	upstream of the construct	CGTATGCTCTACGCACTTCGC	4.4kb
		within the ef1alpha	GGTGCTTTGAGGGGTGAGC	
3'HR	PsΔ TyΔ CtΔ NoΔ	within mCherry	CCTACAAGGCCAAGAAGC	2.7kb
		Within the 3'UTR	CCATGTTCCCATGTGTATCTAC	
Whole locus	PsΔ TyΔ CtΔ NoΔ	Forward	CGTATGCTCTACGCACTTCGC	8.8kb
		Reverse	CCATGTTCCCATGTGTATCTAC	
Control	PsΔ TyΔ CtΔ NoΔ	Forward	TGGAGCAGGAAATAACTGGG	470bp
		Reverse	ACCTGACATAGCGGCTGAAA	

Method/Kits	Mutations	Primer pair	Primer Sequence
Mutagenesis/ QuickChangeII	PsΔ	Primer	GTCTGCCTCATATCCTTCATCCGCTCCTTCGCCATCATTT GTTTC
		antisense	GAAACAAATGATGGCGAAGGAGCGGATGAAGGATATGAG GCAGAC
	TyΔ	Primer	GTAAGAAAAACGGCGGCCTTATCCTTCATCTTCTCCTTCG
		antisense	CGAAGGAGAAGATGAAGGATAAGGCCGCCGTTTTCTTA C
	CtΔ	Primer	GTAAGAAAAACGGCGGCCTTAGCCATCATTTGTTTCAATT CC
		antisense	GGAATTGAAACAAATGATGGCTAAGGCCGCCGTTTTCTT AC

Table 3

***Plasmodium berghei* α1-tubulin mutagenesis QuickChangeII primers sequences**

vein was severed to allow perfusion with HEPES buffer for 5 minutes, during which periodic clamping (10 seconds per minute) of the cut vein was performed. The tubing was subsequently changed to a collagenase solution, and perfusion was held for 7 minutes with repeated clamping. The liver was excised, transferred to a collagenase-containing well, and mechanically dissociated using filter tips and gentle shaking with forceps. Cell suspensions were sequentially collected, pooled, and washed with William's Medium E (WME) by centrifugation at 50g for 2 minutes. The pellet was resuspended in WME, optionally passed through a 70 μ m nylon strainer, and washed twice more. The final pellet was resuspended in WME complete medium (WME supplemented with L-glutamine, penicillin-streptomycin, amphotericin B, insulin-transferrin, and aprotinin), and cells were counted and seeded at densities appropriate for T25 flasks, GB dishes, or 24-well plates. After initial adhesion (1 hour), culture medium was supplemented or exchanged after 2–4 hours.

Immunofluorescence assay (IFA) of fixed primary hepatocytes and HeLa cells

Immunofluorescence was carried out as previously described ([56](#), [86](#), [103](#)). Cells were seeded onto coverslips in culture 24 well-plates and allowed to adhere overnight before infection. Infection was maintained for 24 hours post-infection (hpi), after which cells were fixed using cold methanol at -20°C for 15 minutes and subsequently washed twice with Phosphate buffered saline (PBS). Blocking was performed by incubating cells in 10% FCS in PBS for 20 minutes at room temperature to reduce non-specific binding. Primary antibodies staining was carried out by incubating cells in 10% of FCS diluted in PBS for 1 hour at room temperature, followed by three washes with PBS. Secondary antibodies incubation was performed in 10% of FCS as well for 1 hour at room temperature in the dark. The antibodies used are listed in the enumerated session below. After three additional washes with PBS, nuclei were counterstained with DAPI. Coverslips were then mounted onto slides with DAKO mounting medium and fluorescence imaging was performed using either Nikon W1 or LIPSI and Crest V3 confocal microscope equipped with filters for DAPI (4,6-Diamidin-2-phenylindol), AF488, AF594, and Cy5. Single-stain controls were carried out for every staining to ensure that bleed-through was negligible.

Sporozoite isolation from mosquito salivary glands and haemolymph

Anopheles mosquitoes were fed on *Plasmodium*-infected mice exhibiting a gametocytaemia of 0.2 to 1% and the sporozoite isolations were performed from day 18 post-infection as previously described ([16](#), [61](#)). For salivary gland sporozoites, mosquitoes were anesthetized with chloroform, sterilized in 70% ethanol, and rinsed twice in PBS. The mosquitoes were then dissected, and salivary glands were extracted under a binocular microscope in Iscove's Modified Dulbecco's Medium (IMDM) pooled in 50 μ L of ice-cold IMDM and mechanically homogenized using a pestle (Sigma-Aldrich, Z359947) powered with cordless motor (Sigma-Aldrich, Z359971) as described ([16](#), [61](#)). For hemolymph sporozoite extraction, glass capillaries were heat-pulled to fine tips. The abdominal terminal segment of the mosquito was excised, and hemolymph was flushed from the thorax by PBS perfusion via capillary insertion ([75](#), [104](#)). Hemolymphs extracted were centrifuged (11,000 $\times$ g, 1 min), and pellets were resuspended in PBS or IMDM. Sporozoite quantification was performed using a Neubauer hemacytometer. Midgut, hemolymph, or salivary gland suspensions were mixed 1:1 with RPMI-1640 supplemented 10% FCS, and sporozoites within four corner grids were counted. Total yields (sporozoites/mL and per mosquito) were calculated based on dilution factors and dissection numbers.

Sporozoites IFA and gliding motility assay

To perform the sporozoite immunofluorescence assay (IFA), 12 mm cover glasses (0111530, Paul Marienfeld, Germany) were coated with 0.1 mg/mL poly-D-lysine (A38904-01, Gibco, USA) ([56](#), [105](#)) and incubated overnight at room temperature. The following day, sporozoites were isolated from mosquito midguts, hemolymph, or salivary glands as described above. The extracted

sporozoites were suspended in RPMI-1640 medium to a final volume of 200 μ L in an Eppendorf tube and centrifuged at 1000 rpm for 3 min. The pellet was fixed by resuspension in 200 μ L of cold methanol and incubated for 15 min at -20°C . After fixation, sporozoites were washed twice with PBS (1000 rpm centrifugation) and transferred onto poly-D-lysine-coated cover glasses in a 24-well plate (400024, Bioswisstec, Switzerland). The plate was incubated for 30 min at room temperature to facilitate sporozoite adhesion before proceeding with either the IFA or expansion microscopy protocol.

For the gliding motility assay, 5 μ L of the sporozoite suspension in RPMI-1640 (supplemented with 10% FCS) was added to 96-well glass-bottom plates (P96-1.5H-N, Cellvis, USA) (56 [72](#), 73 [73](#)). The plate was centrifuged (1000 rpm, 1 min) and immediately live imaged using a Leica DM5500 widefield fluorescence microscope.

Expansion microscopy

Expansion microscopy (PS-ExM) was performed as previously established (56 [72](#)–58 [73](#)). Infected cells (HeLa cells, primary hepatocytes and sporozoites) were seeded on 13 mm coverslips in 24-well plates and fixed with either 4% paraformaldehyde (PFA; F8775, Sigma) or cold methanol. The cells were anchored by incubation in PBS containing 0.7% formaldehyde and 1% acrylamide at 37°C for 2 hours. Gelation was then performed by adding 5 μ L each of 10% tetramethylethylenediamine (TEMED; 17919, Thermo Fisher) and 10% ammonium persulfate (APS) to 90 μ L of monomer solution containing 19% sodium acrylate (408220, Sigma), 10% acrylamide (A4058, Sigma), and 0.1% N, N'-methylenebisacrylamide (M1533, Sigma). A 35 μ L of monomer solution was dispensed onto parafilm in a humidity chamber, and coverslips containing the anchored cells were inverted onto droplets. Following 5 minutes incubation on ice to improve gel penetration, gels were polymerized at 37°C for 30 minutes. Gel-embedded samples were transferred to denaturation buffer (200 mM SDS, 200 mM NaCl, 50 mM Tris-HCl, pH 9) in 6-well plates and agitated (200 rpm) at room temperature for 15 minutes, followed by incubation in fresh buffer at 95°C for 1 hour with gentle shaking. Gel expansion was performed in autoclaved MilliQ water (50 mL) for 1 hour with optional buffer exchange. For immunostaining, gels were incubated with primary antibodies in 2% BSA for 2 hours at 37°C , washed with PBS-Tween 0.1%, then labelled with secondary antibodies plus 1 $\mu\text{g/mL}$ DAPI for nuclear staining in 2% BSA for 2 hours at 37°C . Final gel expansion was conducted in MilliQ water containing 0.2% propyl gallate (02370, Sigma) to prevent photobleaching. Expanded samples were mounted on poly-D-lysine-coated (A38904, Gibco USA) glass-bottom dishes with cellular surfaces facing downward and imaged with confocal microscopy.

Antibodies used for IFA and expansion microscopy

The primary antibodies used were: anti-UIS4 chicken (1:1000 IFA/1:500 expansion; Proteogenix), anti-UIS4 rabbit (1:1000 IFA/1:500 expansion; P. Sinnis, Baltimore), polyglutamylation IN105 rabbit (1:1000 IFA/1:500 expansion; AG-25B-0030-C050, AdipoGen Life Sciences), anti-GFP rabbit (1:1000 IFA/1:500 expansion; SP3005P, Origene), anti- α -tubulin guinea pig (1:500 IFA/1:125 expansion; AA345, ABCD), anti- β -tubulin guinea pig (1:500 IFA/1:125 expansion; AA344, ABCD), anti-tyrosinated tubulin mouse (1:1000 IFA/1:500 expansion; T9028, Sigma Aldrich), anti-polyglutamylation GT335 mouse (AG-20B-0020-C100, AdipoGen Life Sciences), anti- γ -tubulin mouse (1:1000 IFA/1:500 expansion; T6557, Sigma Aldrich), and anti-GFP mouse (1:1000 IFA/1:500 expansion; 814460001, Roche).

Secondary antibodies included anti-chicken IgY Alexa Fluor 594 (1:1000 IFA/1:500 expansion; A11042, Invitrogen Molecular Probes), anti-rabbit IgG Alexa Fluor 594 (1:1000 IFA/1:500 expansion; A21207, Life Technologies) and 488 (1:1000 IFA/1:500 expansion; A11034, Sigma Aldrich), anti-rabbit IgG ATTO 647N (1:1000 expansion only; 40839, Sigma Aldrich), anti-guinea pig IgG Alexa Fluor 647 (1:1000 IFA/1:500 expansion; SAB4600033, Sigma Aldrich) and 594 (1:1000 IFA/1:500

expansion; A11076, Invitrogen Molecular Probes), and anti-mouse IgG ATTO 647N (1:1000 expansion only; 50185, Sigma Aldrich) and Alexa Fluor 488 (1:1000 IFA/1:500 expansion; A11001, Invitrogen Molecular Probes).

In vivo assay

To assess the infectivity of parasite mutant sporozoites to hepatocytes, *Anopheles stephensi* mosquitoes infected with different parasite mutants were allowed to feed on anesthetized naïve mice as previously described (Atchou et al., 2025). Three mice per parasite line and five mosquitoes per mouse were used. The blood stage infection progression was monitored daily via flow cytometry, starting from the day of infection until mice tested positive.

Oocyst size measurement

To quantify oocyst size, *Anopheles stephensi* mosquitoes were allowed to feed on anesthetized mice infected with the parasite (Rogerio et al. 2006; Hegge et al., 2009). At 7- and 11-days post-feeding, ten mosquitoes per time point were dissected, and their midguts were extracted and placed in 1× PBS. The midguts were then transferred to a 21-well microscopy slide, covered with a coverslip, and imaged using a Nikon Crest V3 confocal microscope (10× objective). The oocyst area was measured using Fiji software (ImageJ). Each experiment was performed in triplicate for every parasite line.

Liver stage parasite size measurement

To quantify the parasite size during liver stage development, HeLa cells (40,000 cells/well, triplicate) were seeded in 96-well plates and infected 24 hours later with 20,000 sporozoites per well. After a 2-hour incubation at 37°C, infected cells from each well were washed and redistributed into eight wells across two 96 well plates (4 wells/plate) to optimize cell density for imaging (Pino et al., 2017; Atchou et al., 2025; Sturm et al., 2006). Parasite development was analyzed at 6, 24, and 48 hpi using an InCell Analyzer 2000 with 10x objective. The image processing parameters such as kernel size, sensitivity, and size thresholds were optimized in InCell Developer software to distinguish individual parasites from background signals. Parasite growth was assessed based on size measurements, while survival rates were calculated as percentages relative to 6 hpi baseline counts.

Live cell imaging of *Plasmodium* microtubule dynamism

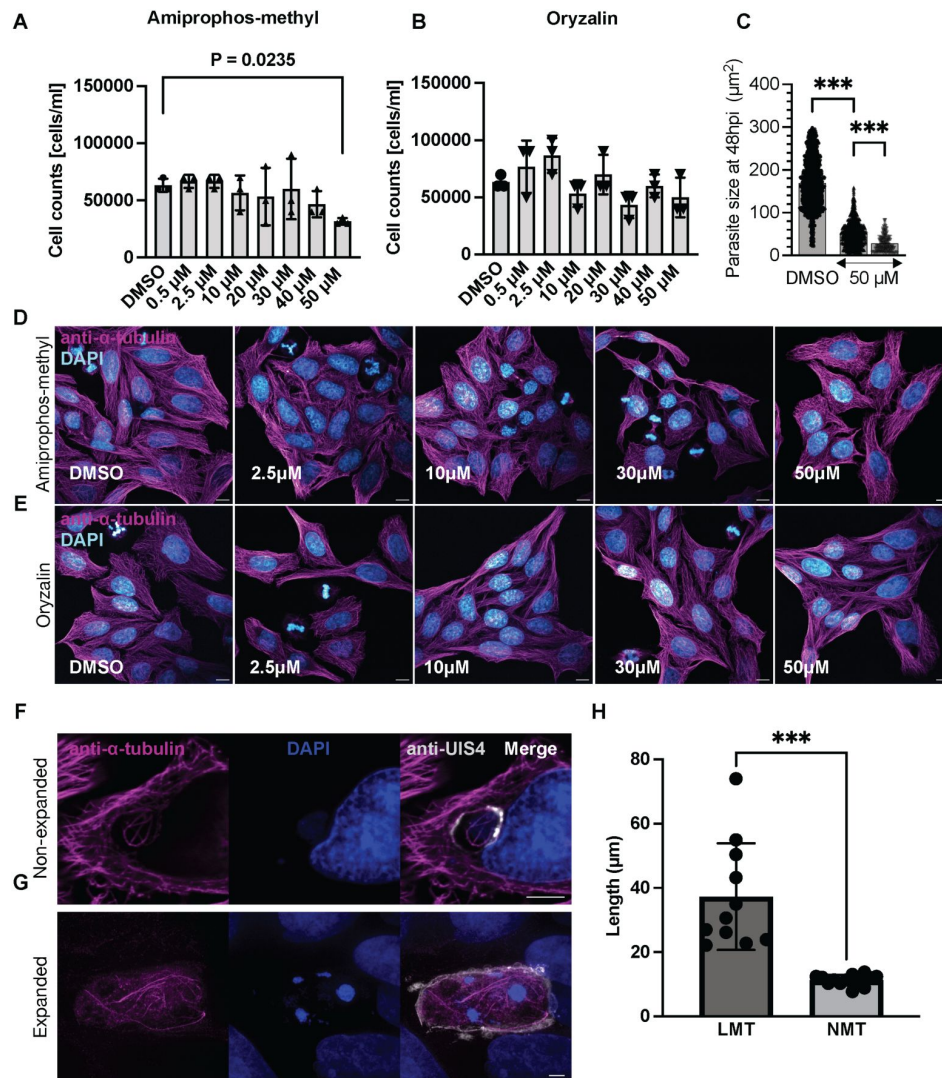
HeLa cells were seeded in a 96-well plate and infected the following day with *P. berghei* salivary gland sporozoites. After a 2-hour incubation at 37°C, cells were washed with MEM medium to remove uninfected parasites and residual mosquito salivary gland materials. The infected cells were then reseeded in a dark 24-well glass-bottom plate (Cellvis) and maintained in fresh cMEM at 37°C with 5% CO₂ (16 [DOI](#), 106 [DOI](#)). Prior to live imaging, cells were incubated for 1 hour in cMEM containing the SPY650-tubulin probe (1:1000 dilution; SC503, Spirochrome, Switzerland). Images were acquired using a Nikon W1 LIPSI spinning disk microscope with a 100X objective. Marked infected HeLa cells were imaged at defined time points to generate time-lapse movies capturing tubulin dynamics and parasite development.

Image processing and data analysis

All the live cell imaging, expansion microscopy, and IFA images in this study were acquired with Nikon W1 LIPSI and Crest V3, both in spinning disk mode. The microscope images were processed and analyzed using FIJI software, Huygens deconvolution Professional version (Huygens Remote Manager v3.8, Scientific Volume Imaging, The Netherlands), or, when specified, the 3D and 4D IMARIS software (Imaris 10.2.0; June 18, 2024). The 3D/4D image analysis software was used to segment the parasite LSPMB, hemi-spindle, and poles into either isospots or isosurfaces, as previously described (16 [DOI](#), 56 [DOI](#)). The counts, volumes, and coefficients of the isospots or

isosurfaces were retrieved in .csv files, which were then imported into GraphPad Prism for statistical analysis. All data were analyzed using GraphPad Prism version 10.4.1 for macOS (GraphPad Software, San Diego, California, USA). Statistical comparisons were performed using one-way ANOVA with Dunn's multiple comparisons test (* $p \leq 0.05$, $p \leq 0.01$, * $p \leq 0.001$). All experiments in the study were carried out in triplicate.

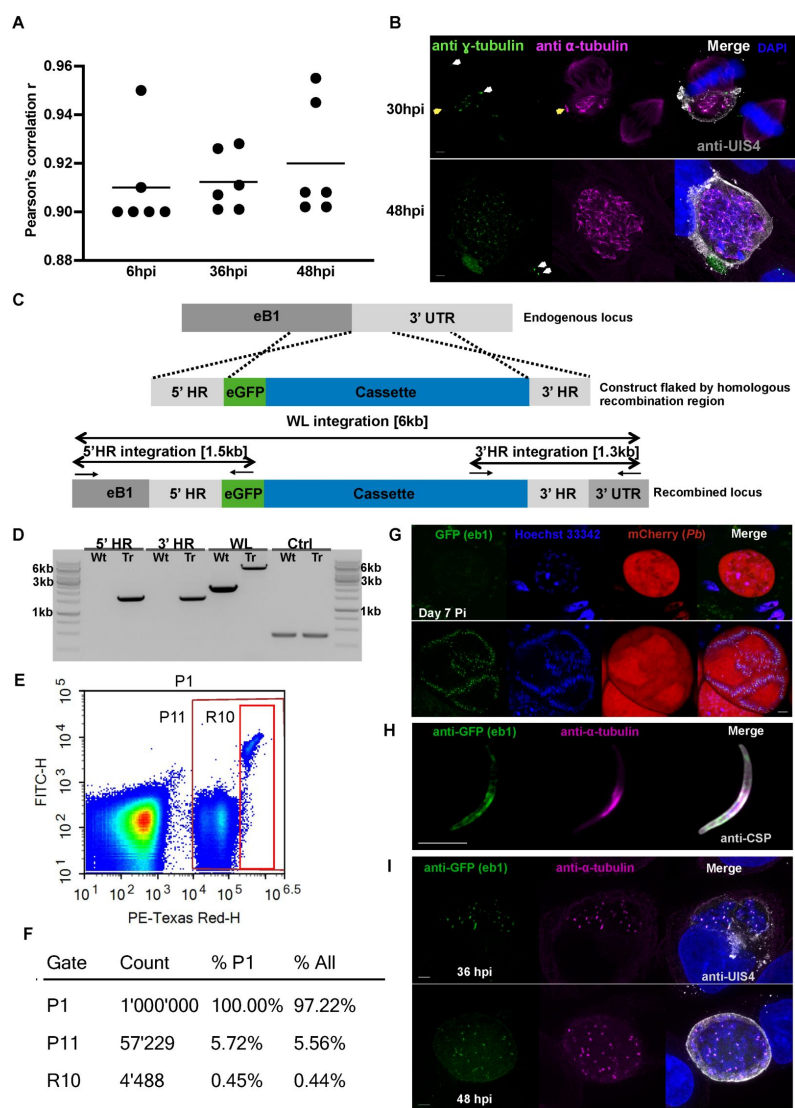
Supplementary figures



S1 Figure

Effect of *Plasmodium*-specific microtubule inhibitors on HeLa cell.

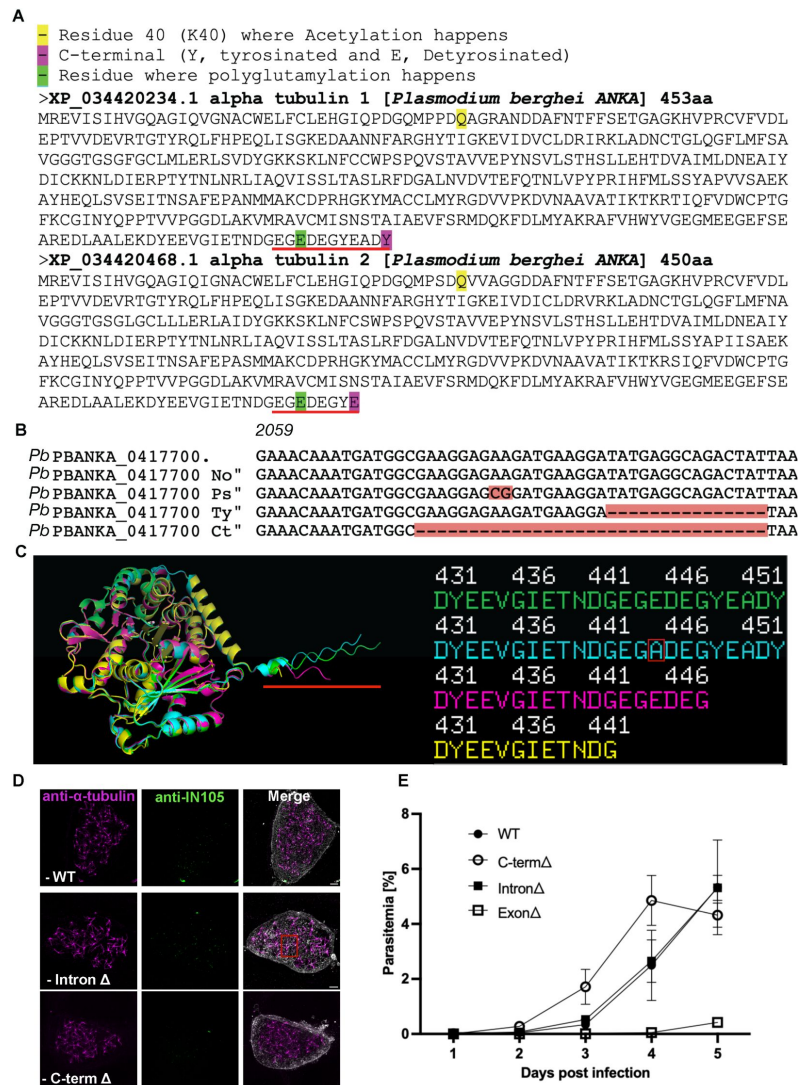
HeLa cells were seeded in 6-well plates and treated the following day with various concentrations of microtubule-targeting drugs. Media were replaced daily with fresh drug-supplemented medium. After five days, surviving cells were detached and counted. **(A)** Effect of amiprophos-methyl (AMP) on HeLa cell viability. A significant reduction in cell count was observed only at 50 μ M compared to the DMSO control. **(B)** effect of the oryzalin on HeLa cells viability. No significant differences in cell count were observed at any tested concentrations tested relative to DMSO. **(C)** represent the parasite size at 48 hpi in infected HeLa cells treated with oryzalin and AMP at 50 μ M. oryzalin showed a significant inhibitory effect on parasite growth compared to AMP and DMSO control. Statistical comparison tests were performed using a one-way ANOVA test (* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$). **(D and E)** Maximum projection confocal IFA z-stacks of HeLa cells treated with amiprophos-methyl (D) and oryzalin (E) tested at different concentrations, respectively. Cells were fixed with cold and stained with anti- α -tubulin (magenta) and DAPI (blue). **(F)** Confocal IFA z-projection of HeLa cells treated with low-dose oryzalin (0.5 μ M), showing unusually elongated microtubule structures. **(G)** PS-ExM confocal z-stack of oryzalin-treated cells showing the very long LSPMB and hemi-spindles. **(H)** Quantification of LSPMB and hemi-spindle lengths. Oryzalin-treated microtubules were on average four times longer than those in DMSO-treated controls. Each dot represents a measurement from a single cell. LMT, Long microtubule; NMT, normal microtubule. Statistical comparison tests were performed using two tailed unpaired t tests (* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$).



S2 Figure

Plasmodium end-binding protein 1 (eB1) is expressed in mosquito and liver stag.

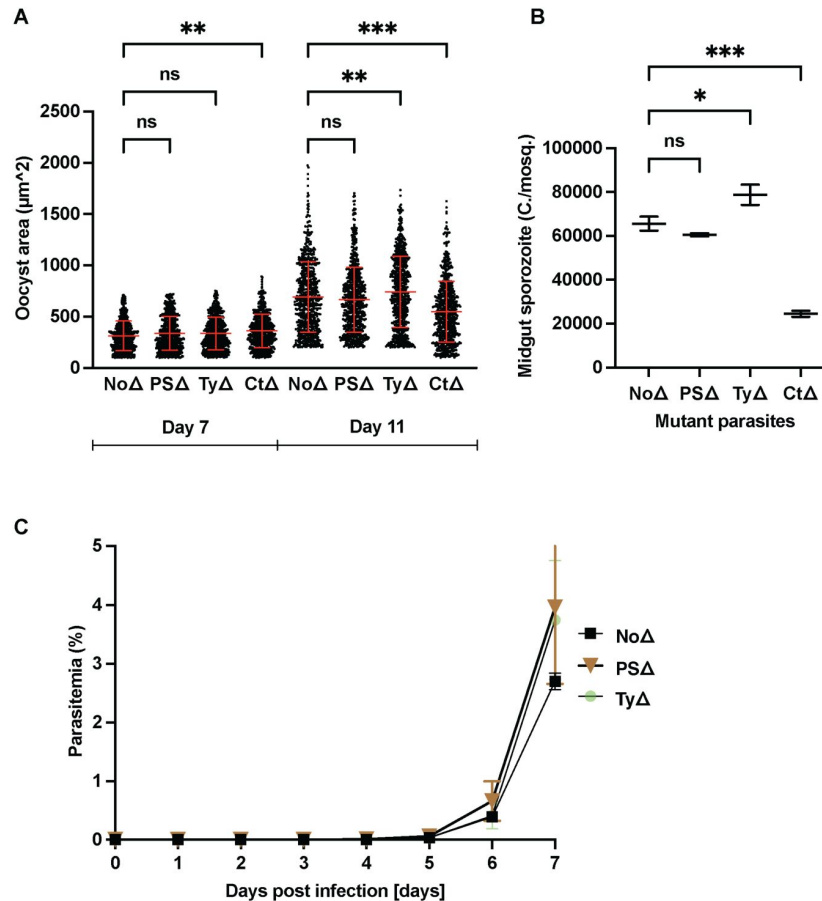
(A) Pearson correlation coefficients between polyglutamylated LSPMB and α -tubulin LSPMB at 6, 36, and 48 hpi during liver stage development. On average, 92% of polyglutamylated tubulin colocalized with the LSPMB. **(B)** Maximum projection of PS-ExM confocal z-stacks of infected HeLa cells fixed at 30 and 48 hpi. Cells were stained with anti- γ -tubulin (green), anti- α -tubulin (magenta), anti-UIS4 (grey, PVM), and DAPI (blue, nuclei). White arrows indicate host centrosomes; yellow arrows highlight the LSPMB. Parasite hemi-spindle poles were also detected with γ -tubulin. **(C)** Schematic of the double homologous recombination construct used to tag *Plasmodium* eB1 (PBANKA_0405600) with eGFP at the C-terminus. The expression cassette (blue box) includes 3'UTR *PbDHFR*, EF1 α promoter, hDHFR, yeast *fcu*, and the 3'UTR of PBANKA_0405600, flanked by the 5'HR (eB1 coding region) and 3'UTR. The construct was transfected into the PbmCherry 1868 mother line. **(D)** Integration PCR confirming correct insertion of the eGFP-tagged eB1 cassette. PCR product sizes: 5'HR = 1.5 kb, 3'HR = 1.3 kb, whole locus (WL) = 6 kb. Primer sequences listed in **Table 1**. Lane labels: Tr (transgenic), WT (wild type), ctrl (control). Ladder: 1 kb DNA marker. **(E)** Flow cytometry profile of erythrocytes infected with eB1-GFP-expressing parasites. Gate P1 includes all analyzed cells; P11 identifies mCherry-positive (PE-Texas Red-H) infected erythrocytes; R10 identifies GFP+/mCherry+ gametocytes. **(F)** Summary table showing parasitemia percentages across gates defined in (E). **(G)** Representative confocal live imaging of *P. berghei* eB1-GFP oocysts at 7 and 12 days post-infection (dpi), showing eB1-GFP (green), parasite mCherry (red), and Hoechst-stained nuclei (blue). Scale bar: 5 μ m. **(H)** Confocal IFA image of *P. berghei* eB1-GFP sporozoites stained with anti-GFP (green), anti- α -tubulin (magenta), and anti-circumsporozoite protein (CSP, grey). **(I)** Maximum projection of PS-ExM confocal z-stacks of HeLa cells infected with *P. berghei* eB1-GFP parasites at 36 and 48 hpi. Cells were stained with anti-GFP (green), anti- α -tubulin (magenta), and anti-UIS4 (grey). Scale bar: 10 μ m.



S3 Figure

***Plasmodium* α-tubulin proteins and mutant parasite characterization.**

(A) Comparison of *Plasmodium* α1- and α2-tubulin protein sequences highlighting predicted post-translational modification (PTM) sites. Polyglutamylation site E445 is conserved in both isoforms (green). α1-tubulin contains a predicted tyrosination site at Y453 (magenta), while α2-tubulin has a predicted detyrosination site at E450 (magenta). Neither isoform has the conserved acetylation site K40 (yellow). C-terminal PTM regions are underlined in red. (B) Sequence alignment of PBANKA_0417700 gene showing mutations introduced in the tubulin mutant parasites. Red highlights indicate substitutions or deletions. Dashes (—) indicate deleted nucleotides. Ps": glutamine (E) substituted with alanine (A) at the polyglutamylation site; Ty": deletion of the C-terminal tyrosination/detyrosination motif; Ct": deletion of the entire C-terminal PTM region; No": unmodified construct control. (C) Structural comparison of α1-tubulin mutant proteins using AlphaFold 3 predictions, aligned with PyMOL (Version 2.6.0). Green: wild type (No"); cyan: Ps" mutant; magenta: Ty" mutant; and yellow: Ct" mutant. All predicted structures retained overall folding. (D) Maximum projection of confocal z-stacks of HeLa cells infected with mutant parasites (adapted from Freedy et al.). Microtubules were stained with anti-α-tubulin (magenta) and polyglutamylated tubulin with anti-IN105 (green). Nuclei were stained with DAPI (blue); the parasitophorous vacuole membrane (PVM) was stained with anti-UIS4 (grey). IntronΔ: line with deleted introns in the *Plasmodium* α1-tubulin gene; C-termΔ: line with deletion of C-terminal residues 540ADY. (E) In vivo infection kinetics in mice infected with mutant parasites. Parasitemia was measured daily from day 1 to day 5 post-infection. No significant differences were observed between mutant lines (intronΔ and C-termΔ) and wild type (WT). ExonΔ: mutant line in which α1-tubulin exons were replaced with α2-tubulin introns. This line failed to produce salivary gland sporozoites. Liver stage development in this line was delayed by 4 days due to hemolymph sporozoites used. Experiments were performed in triplicate.



S4 Figure

Plasmodium α -tubulin mutant parasite oocyst formation and sporozoite infectivity.

(A) Comparison of oocyst area at day 7 and 11 post-infection. No significant difference was observed between the WT (No Δ) and the PS Δ or Ty Δ mutants at day 7 post-infection. However, by day 11, the Ct Δ mutant exhibited a significantly reduced oocyst size compared to the WT. **(B)** Normalized midgut sporozoites count per mosquito (C./mosq.) at day 14 post-feeding. No significant difference in sporozoite numbers was observed between No Δ and PS Δ , whereas Ct Δ showed a drastic reduction. In contrast, Ty Δ displayed both a significantly larger oocyst size at day 11 and a higher sporozoite count at day 14 compared to the WT. **(C)** Blood parasitemia levels in mice infected with mutant parasites, measured from day 0 to day 7 post-infection. All mutants (PS Δ and Ty Δ) established infections and progressed similarly to the WT (No Δ). PS Δ : glutamine-to-alanine substitution at polyglutamylation site; Ty Δ : deletion of C-terminal tyrosination/detyrosination residues; Ct Δ : complete deletion of PBANKA_0417700 α -tubulin C-terminal PTM residues; No Δ : control construct with no mutation. The experiments were conducted in triplicate. Statistical multiple comparisons were performed using one-way ANOVA with Dunnett's multiple comparisons test (* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$).

Acknowledgements

The authors gratefully acknowledge the Microscopy Imaging Center (MIC) at the University of Bern, Switzerland, for their technical support and access to microscopy equipment. K.A. acknowledges the Swiss Confederation for the award of a Government Excellence Scholarship. *This work was funded by the Swiss National Science Foundation (SNSF) (grant number [310030_212795](#)) and an Multidisciplinary Center for Infectious Diseases (MCID), University of Bern, Switzerland, grant to Volker Heussler (MA-09).*

Additional files

Movie 1. Time-lapse live imaging of *P. berghei* salivary gland sporozoites expressing GFP cytosolically. Sporozoites were isolated and incubated with tubulin tracker SPY650. Microtubules are shown in magenta; sporozoites are shown in green. Scale bar: 5 μ m. [🔗](#)

Movie 2. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies show the development and dynamics of the LSPMB during liver stage infection. Microtubules are shown in magenta; parasites are shown in green. Imaging from 10 hpi to 25 hpi. Scale bar: 5 μ m. [🔗](#)

Movie 3. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies show the development and dynamics of the LSPMB during liver stage infection. Microtubules are shown in magenta; parasites are shown in green. Imaging from 30 hpi to 35 hpi. Scale bar: 5 μ m. [🔗](#)

Movie 4. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies show the development and dynamics of the LSPMB during liver stage infection. Microtubules are shown in magenta; parasites are shown in green. Imaging from 37 hpi to 49 hpi. Scale bar: 5 μ m. [🔗](#)

Movie 5. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies highlight the high dynamicity of the LSPMB during liver stage development, starting from early schizogony (~30 hpi). Microtubules are shown in magenta; parasites are shown in green. Scale bar: 5 μ m. [🔗](#)

Movie 6. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies highlight the high dynamicity of the LSPMB during liver stage development, starting from early schizogony (~30 hpi). Microtubules are shown in magenta; parasites are shown in green. Scale bar: 5 μ m. [🔗](#)

Movie 7. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies highlight the high dynamicity of the LSPMB during liver stage development, starting from early schizogony (~30 hpi). Microtubules are shown in magenta; parasites are shown in green. Scale bar: 5 μ m. [🔗](#)

Movie 8. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies reveal the interaction between the LSPMB and the parasite's hemi-spindles and spindle poles during liver stage development. Microtubules are shown in magenta; parasites are shown in green. LSPMB (white arrow) localizes at the periphery of the parasite and connects to hemi-spindles. Scale bar: 5 μm . [🔗](#)

Movie 9. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies reveal the interaction between the LSPMB and the parasite's hemi-spindles and spindle poles during liver stage development. Microtubules are shown in magenta; parasites are shown in green. LSPMB (white arrows) connects to the parasite's hemi-spindle poles. Scale bar: 5 μm . [🔗](#)

Movie 10. Time-lapse live imaging of HeLa cells infected with *P. berghei* liver stage parasites expressing cytosolic GFP. Infected cells were incubated with tubulin tracker SPY650 for one hour prior to imaging. The movies reveal the interaction between the LSPMB and the parasite's hemi-spindles and spindle poles during liver stage development. Microtubules are shown in magenta; parasites are shown in green. Shrinkage of the LSPMB during the liver stage development (ROI marked by a white arrow). Scale bar: 5 μm . [🔗](#)

Supplementary Movie 1. 3D projection of PS-ExM images showing LSPMB connected to the hemi-spindle pole in *P. berghei*-infected primary hepatocytes. Fixed and expanded primary hepatocytes were stained with anti-IN105 (polyglutamylated tubulin, green), anti- α -tubulin (magenta), and anti-UIS4 (PVM, grey). The movie shows the LSPMB in contact with the parasite's hemi-spindle pole. Image analysis was performed using IMARIS 3D and 4D visualization software. [🔗](#)

Supplementary Movie 2. *Plasmodium berghei* salivary gland (SG) sporozoite motility of *P. berghei* No Δ (control), Ps Δ , and Ty Δ mutants, as well as hemolymph sporozoites of *P. berghei* No Δ and Ct Δ mutants. Scale bar: 5 μm [🔗](#)

References

1. Wambani J, Okoth P. (2022) **Impact of Malaria Diagnostic Technologies on the Disease Burden in the Sub-Saharan Africa** <https://onlinelibrary.wiley.com/doi/10.1155/2022/7324281>
2. Poespoprodjo JR, Douglas NM, Ansong D, Kho S, Anstey NM. (2023) **Malaria** *Lancet* **402**:2328–45 [PubMed](#) | [Google Scholar](#)
3. Varo R, Chaccour C, Bassat Q (2020) **Update on malaria** *Med Clin (Barc)* **155**:395–402 [PubMed](#) | [Google Scholar](#)
4. Venkatesan P (2025) **WHO world malaria report 2024** *Lancet Microbe* **6**:101073 <https://www.thelancet.com/action/showFullText?pii=S2666524725000011> | [Google Scholar](#)
5. Tavares J, Formaglio P, Thiberge S, Mordelet E, Van Rooijen N, Medvinsky A, et al. (2013) **Role of host cell traversal by the malaria sporozoite during liver infection** *Journal of Experimental Medicine* **210**:905–15 <https://www.jem.org/cgi/doi/10.1084/jem.20121130> | [Google Scholar](#)
6. Bindschedler A, Wacker R, Egli J, Eickel N, Schmuckli-Maurer J, Franke-Fayard BM, et al. (2021) **Plasmodium berghei sporozoites in nonreplicative vacuole are eliminated by a PI3P-mediated autophagy-independent pathway** *Cell Microbiol* **23** [PubMed](#) | [Google Scholar](#)
7. Marques-da-Silva C, Schmidt-Silva C, Kurup SP (2024) **Hepatocytes and the art of killing Plasmodium softly** *Trends Parasitol* **40**:466–76 <https://www.cell.com/action/showFullText?pii=S1471492224000862> | [Google Scholar](#)
8. Prado M, Eickel N, De Niz M, Heitmann A, Agop-Nersesian C, Wacker R, et al. (2015) **Long-term live imaging reveals cytosolic immune responses of host hepatocytes against Plasmodium infection and parasite escape mechanisms** *Autophagy* **11**:1561–79 [PubMed](#) | [Google Scholar](#)
9. Zanghi G, Vaughan AM (2021) **Plasmodium vivax pre-erythrocytic stages and the latent hypnozoite** *Parasitol Int* **85**:102447 [Google Scholar](#)
10. Prudêncio M, Rodriguez A, Mota MM (2006) **The silent path to thousands of merozoites: the Plasmodium liver stage** *Nature Reviews Microbiology* **4**:849–56 <https://www.nature.com/articles/nrmicro1529> | [Google Scholar](#)
11. Roques M, Bindschedler A, Beyeler R, Heussler VT (2023) **Same, same but different: Exploring Plasmodium cell division during liver stage development** *PLoS Pathog* **19**:e1011210 [Google Scholar](#)
12. Shears MJ, Sekhar Nirujogi R, Swearingen KE, Renuse S, Mishra S, Jaipal Reddy P, et al. (2019) **Proteomic Analysis of Plasmodium Merosomes: The Link between Liver and Blood Stages in Malaria** *J Proteome Res* **18**:3404–18 <https://pubs.acs.org/doi/full/10.1021/acs.jproteome.9b00324> | [Google Scholar](#)
13. De Niz M, Heussler VT (2018) **Rodent malaria models: insights into human disease and parasite biology** *Curr Opin Microbiol* **46**:93–101 [PubMed](#) | [Google Scholar](#)

14. Sturm A, Amino R, Van De Sand C, Regen T, Retzlaff S, Rennenberg A, et al. (1979) **Manipulation of host hepatocytes by the malaria parasite for delivery into liver sinusoids** *Science* **313**:1287–90 <https://www.science.org/doi/10.1126/science.1129720> | [Google Scholar](#)
15. Heussler V, Spielmann T, Frischknecht F, Gilberger T (2016) **Plasmodium** *Molecular Parasitology: Protozoan Parasites and their Molecules* :241–84 https://link.springer.com/chapter/10.1007/978-3-7091-1416-2_9 | [Google Scholar](#)
16. Atchou K, Caldelari R, Roques M, Schmuckli-Maurer J, Beyeler R, Heussler V. (2025) **Expanding the fluorescent toolkit: Blue fluorescent protein-expressing Plasmodium berghei for enhanced multiplex microscopy** *PLoS One* **20**:e0308055 [PubMed](#) | [Google Scholar](#)
17. Stanway RR, Bushell E, Chiappino-Pepe A, Roques M, Sanderson T, Franke-Fayard B, et al. (2019) **Genome-Scale Identification of Essential Metabolic Processes for Targeting the Plasmodium Liver Stage** *Cell* **179**:1112–1128 [PubMed](#) | [Google Scholar](#)
18. Meibalan E, Marti M (2017) **Biology of Malaria Transmission** *Cold Spring Harb Perspect Med* **7** [PubMed](#) | [Google Scholar](#)
19. Frischknecht F, Baldacci P, Martin B, Zimmer C, Thiberge S, Olivo-Marin JC, et al. (2004) **Imaging movement of malaria parasites during transmission by Anopheles mosquitoes** *Cell Microbiol* **6**:687–94 <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1462-5822.2004.00395.x> | [Google Scholar](#)
20. Vanderberg JP (1975) **Development of infectivity by the Plasmodium berghei sporozoite** *Journal of Parasitology* **61**:43–50 [Google Scholar](#)
21. Vanderberg jerome P (1974) **Studies on the Motility of Plasmodium Sporozoites*** *J Protozool* **21**:527–37 <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1550-7408.1974.tb03693.x> | [Google Scholar](#)
22. Frischknecht F, Matuschewski K (2017) **Plasmodium Sporozoite Biology** *Cold Spring Harb Perspect Med* **7**:a025478 [Google Scholar](#)
23. Mitchison T, Kirschner M (1984) **Dynamic instability of microtubule growth** *Nature* **312**:237–42 <https://www.nature.com/articles/312237a0> | [Google Scholar](#)
24. Janke C, Magiera MM (2020) **The tubulin code and its role in controlling microtubule properties and functions** *Nat Rev Mol Cell Biol* **21**:307–26 [Google Scholar](#)
25. Nogales E (2001) **Structural insights into microtubule function** *Annu Rev Biophys Biomol Struct* **30**:397–420 <https://www.annualreviews.org/content/journals/10.1146/annurev.biophys.30.1.397> | [Google Scholar](#)
26. Roll-Mecak A (2020) **The Tubulin Code in Microtubule Dynamics and Information Encoding** *Dev Cell* **54**:7–20 [Google Scholar](#)
27. Bodakuntla S, Jijumon AS, Villablanca C, Gonzalez-Billault C, Janke C (2019) **Microtubule-Associated Proteins: Structuring the Cytoskeleton** *Trends Cell Biol* **29**:804–19 [Google Scholar](#)
28. Lee G (1993) **Non-motor microtubule-associated proteins** *Curr Opin Cell Biol* **5**:88–91 [Google Scholar](#)

29. Peng N, Nakamura F (2023) **Microtubule-associated proteins and enzymes modifying tubulin** *Cytoskeleton* **80**:60–76 <https://onlinelibrary.wiley.com/doi/full/10.1002/cm.21748> | [Google Scholar](#)
30. Peng N, Nakamura F (2023) **Microtubule-associated proteins and enzymes modifying tubulin** *Cytoskeleton (Hoboken)* **80**:60–76 [PubMed](#) | [Google Scholar](#)
31. Fennell BJ, Al-shatr ZA, Bell A (2008) **Isotype expression, post-translational modification and stage-dependent production of tubulins in erythrocytic Plasmodium falciparum** *Int J Parasitol* **38**:527–39 [Google Scholar](#)
32. Rawlings DJ, Fujioka H, Fried M, Keister DB, Aikawa M, Kaslow DC (1992) **α -Tubulin II is a male-specific protein in Plasmodium falciparum** *Mol Biochem Parasitol* **56**:239–50 [Google Scholar](#)
33. Arregui C, Busciglio J, Caceres A, Barra HS (1991) **Tyrosinated and detyrosinated microtubules in axonal processes of cerebellar macroneurons grown in culture** *J Neurosci Res* **28**:171–81 [PubMed](#) | [Google Scholar](#)
34. Webster DR, Gundersen GG, Bulinski JC, Borisy GG (1987) **Differential turnover of tyrosinated and detyrosinated microtubules** *Proc Natl Acad Sci U S A* **84**:9040–4 [PubMed](#) | [Google Scholar](#)
35. Sanyal C, Pietsch N, Ramirez Rios S, Peris L, Carrier L, Moutin MJ (2023) **The detyrosination/re-tyrosination cycle of tubulin and its role and dysfunction in neurons and cardiomyocytes** *Semin Cell Dev Biol* **137**:46–62 [PubMed](#) | [Google Scholar](#)
36. Girão H, Macário-Monteiro J, Figueiredo AC, Silva E Sousa R, Doria E, Demidov V, et al. (2024) **α -tubulin detyrosination fine-tunes kinetochore-microtubule attachments** *Nat Commun* [PubMed](#) | [Google Scholar](#)
37. Gagnon C, White D, Cosson J, Huitorel P, Eddé B, Desbruyères E, et al. (1996) **The polyglutamylated lateral chain of alpha-tubulin plays a key role in flagellar motility** *J Cell Sci* **109**:1545–53 <https://doi.org/10.1242/jcs.109.6.1545> | [Google Scholar](#)
38. Edde B, Rossier J, Le Caer JP, Prome JC, Desbruyeres E, Gros F, et al. (1992) **Polyglutamylated alpha-tubulin can enter the tyrosination/detyrosination cycle** *Biochemistry* **31**:403–10 [PubMed](#) | [Google Scholar](#)
39. Ruse CI, Chin HG, Pradhan S. (2022) **Polyglutamylation: biology and analysis** *Amino Acids* **54**:529–42 [PubMed](#) | [Google Scholar](#)
40. Zheng P, Obara CJ, Szczesna E, Nixon-Abell J, Mahalingan KK, Roll-Mecak A, et al. (2022) **ER proteins decipher the tubulin code to regulate organelle distribution** *Nature* **601**:132–8 [PubMed](#) | [Google Scholar](#)
41. Neff NF, Thomas JH, Grisafi P, Botstein D (1983) **Isolation of the beta-tubulin gene from yeast and demonstration of its essential function in vivo** *Cell* **33**:211–9 [PubMed](#) | [Google Scholar](#)
42. Baum P, Thorner J, Honig L (1978) **Identification of tubulin from the yeast Saccharomyces cerevisiae** *Proc Natl Acad Sci U S A* **75**:4962–6 [Google Scholar](#)
43. Hirst WG, Facht D, Kuropka B, Weise C, Saliba KJ, Reber S (2022) **Purification of functional Plasmodium falciparum tubulin allows for the identification of parasite-specific microtubule inhibitors** *Current Biology* **32**:919–926 [Google Scholar](#)

44. Holloway SP, Gerousis M, Delves CJ, Sims PFG, Scaife JG, Hyde JE (1990) **The tubulin genes of the human malaria parasite *Plasmodium falciparum*, their chromosomal location and sequence analysis of the α -tubulin II gene** *Mol Biochem Parasitol* **43**:257–70 [Google Scholar](#)
45. Fennell BJ, Naughton JA, Dempsey E, Bell A (2006) **Cellular and molecular actions of dinitroaniline and phosphorothioamidate herbicides on *Plasmodium falciparum*: tubulin as a specific antimalarial target** *Mol Biochem Parasitol* **145**:226–38 [PubMed](#) | [Google Scholar](#)
46. Dow GS, Armson A, Boddy MR, Itenge T, McCarthy D, Parkin JE, et al. (2002) **Plasmodium: assessment of the antimalarial potential of trifluralin and related compounds using a rat model of malaria, *Rattus norvegicus*** *Exp Parasitol* **100**:155–60 [PubMed](#) | [Google Scholar](#)
47. Dempsey E, Prudêncio M, Fennell BJ, Gomes-Santos CS, Barlow JW, Bell A (2013) **Antimitotic herbicides bind to an unidentified site on malarial parasite tubulin and block development of liver-stage *Plasmodium* parasites** *Mol Biochem Parasitol* **188**:116–27 [PubMed](#) | [Google Scholar](#)
48. Kooij TWA, Franke-Fayard B, Renz J, Kroeze H, Van Dooren MW, Ramesar J, et al. (2005) ***Plasmodium berghei* alpha-tubulin II: a role in both male gamete formation and asexual blood stages** *Mol Biochem Parasitol* **144**:16–26 [PubMed](#) | [Google Scholar](#)
49. Holloway SP, Sims PFG, Delves CJ, Scaife JG, Hyde JE (1989) **Isolation of alpha-tubulin genes from the human malaria parasite, *Plasmodium falciparum*: sequence analysis of alpha-tubulin** *Mol Microbiol* **3**:1501–10 [PubMed](#) | [Google Scholar](#)
50. Spreng B, Fleckenstein H, Kübler P, Di Biagio C, Benz M, Patra P, et al. (2019) **Microtubule number and length determine cellular shape and function in *Plasmodium*** *EMBO J* **38** <https://www.embopress.org/doi/10.15252/embj.2018100984> | [Google Scholar](#)
51. Zhang G, Niu G, Hooker-Romera D, Shabani S, Ramelow J, Wang X, et al. (2023) **Targeting plasmodium α -tubulin-1 to block malaria transmission to mosquitoes** *Front Cell Infect Microbiol* **13**:1132647 [Google Scholar](#)
52. Valenstein ML, Roll-Mecak A (2016) **Graded Control of Microtubule Severing by Tubulin Glutamylation** *Cell* **164**:911–21 <https://www.cell.com/action/showFullText?pii=S0092867416000593> | [Google Scholar](#)
53. Bonnet C, Boucher D, Lazereg S, Pedrotti B, Islam K, Denoulet P, et al. (2001) **Differential Binding Regulation of Microtubule-associated Proteins MAP1A, MAP1B, and MAP2 by Tubulin Polyglutamylation** *Journal of Biological Chemistry* **276**:12839–48 <https://www.jbc.org/action/showFullText?pii=S0021925819343935> | [Google Scholar](#)
54. Bertiaux E, Balestra AC, Bournonville L, Louvel V, Maco B, Soldati-Favre D, et al. (2021) **Expansion microscopy provides new insights into the cytoskeleton of malaria parasites including the conservation of a conoid** *PLoS Biol* **19** [PubMed](#) | [Google Scholar](#)
55. Hentzschel F, Binder AM, Dorner LP, Herzel L, Nuglisch F, Sema M, et al. (2025) **Microtubule inner proteins of *Plasmodium* are essential for transmission of malaria parasites** *Proceedings of the National Academy of Sciences* **122**:e2421737122 <https://www.pnas.org/doi/abs/10.1073/pnas.2421737122> | [Google Scholar](#)
56. Atchou K, Berger BM, Heussler V, Ochsenreiter T (2023) **Pre-gelation staining expansion microscopy for visualisation of the *Plasmodium* liver stage** *J Cell Sci* **136**:jcs261377 [Google Scholar](#)

57. Liffner B, Absalon S (2024) **Expansion microscopy of apicomplexan parasites** *Mol Microbiol* **121**:619–35 [PubMed](#) | [Google Scholar](#)
58. Liffner B, Diaz AKC, Blauwkamp J, Anaguano D, Frolich S, Muralidharan V, et al. (2023) **Atlas of Plasmodium falciparum intraerythrocytic development using expansion microscopy** *Elife* **12**:RP88088 [PubMed](#) | [Google Scholar](#)
59. Ferreira JL, Pražák V, Vasishtan D, Siggel M, Hentzschel F, Binder AM, et al. (2023) **Variable microtubule architecture in the malaria parasite** *Nat Commun* **14**:1216 [Google Scholar](#)
60. Aguayo-Ortiz R, Dominguez L (2022) **Unveiling the Possible Oryzalin-Binding Site in the α -Tubulin of Toxoplasma gondii** *ACS Omega* **7**:18434–42 <https://pubs.acs.org/doi/full/10.1021/acsomega.2c00729> | [Google Scholar](#)
61. Beyeler R, Jordan M, Dorner L, He B, Cyrklaff M, Roques M, et al. (2024) **Putative prefoldin complex subunit 5 of Plasmodium berghei is crucial for microtubule formation and parasite development in the mosquito** *Mol Microbiol* **121**:481–96 [PubMed](#) | [Google Scholar](#)
62. Roques M, Stanway RR, Rea EI, Markus R, Brady D, Holder AA, et al. (2019) **Plasmodium centrin PbCEN-4 localizes to the putative MTOC and is dispensable for malaria parasite proliferation** *Biol Open* **8**:bio036822 [Google Scholar](#)
63. Zeeshan M, Shilliday F, Liu T, Abel S, Mourier T, Ferguson DJP, et al. (2019) **Plasmodium kinesin-8X associates with mitotic spindles and is essential for oocyst development during parasite proliferation and transmission** *PLoS Pathog* **15**:e1008048 [Google Scholar](#)
64. Yang S, Cai M, Huang J, Zhang S, Mo X, Jiang K, et al. (2023) **EB1 decoration of microtubule lattice facilitates spindle-kinetochore lateral attachment in Plasmodium male gametogenesis** *Nature Communications* **14**:1–20 <https://www.nature.com/articles/s41467-023-38516-3> | [Google Scholar](#)
65. Mauer S, Camargo N, Abatiyow BA, Gargaro OR, Kappe SHI, Kumar S (2023) **Plasmodium microtubule-binding protein EB1 is critical for partitioning of nuclei in male gametogenesis** *mBio* **14**:e00822–23 [Google Scholar](#)
66. Magiera MM, Janke C (2013) **Investigating Tubulin Posttranslational Modifications with Specific Antibodies** *Methods Cell Biol* **115**:247–67 [Google Scholar](#)
67. Oakley BR (2000) **gamma-Tubulin** *Curr Top Dev Biol* **49**:27–54 [PubMed](#) | [Google Scholar](#)
68. Maessen S, Wesseling JG, Smits MA, Konings RNH, Schoenmakers JGG (1993) **The γ -tubulin gene of the malaria parasite Plasmodium falciparum** *Mol Biochem Parasitol* **60**:27–35 [Google Scholar](#)
69. Haase R, Puthenpurackal A, Maco B, Guérin A, Soldati-Favre D (2024) **γ -tubulin complex controls the nucleation of tubulin-based structures in Apicomplexa** *Mol Biol Cell* **35**:ar121 [Google Scholar](#)
70. Barik S, Andrews J (2024) **Host-Parasite Interactions in Toxoplasma gondii-Infected Cells: Roles of Mitochondria, Microtubules, and the Parasitophorous Vacuole** . *Int J Mol Sci* **25**:13459 [Google Scholar](#)
71. Burda PC, Caldelari R, Heussler VT (2017) **Manipulation of the Host Cell Membrane during Plasmodium Liver Stage Egress** *mBio* **8** [PubMed](#) | [Google Scholar](#)

72. Hopp CS, Chiou K, Ragheb DRT, Salman AM, Khan SM, Liu AJ, et al. (2015) **Longitudinal analysis of plasmodium sporozoite motility in the dermis reveals component of blood vessel recognition** *eLife* **4** [Google Scholar](#)
73. Beyer K, Kracht S, Kehrer J, Singer M, Klug D, Frischknecht F (2021) **Limited Plasmodium sporozoite gliding motility in the absence of TRAP family adhesins** *Malar J* **20**:1–12 <https://malariajournal.biomedcentral.com/articles/10.1186/s12936-021-03960-3> | [Google Scholar](#)
74. Klug D, Frischknecht F (2017) **Motility precedes egress of malaria parasites from oocysts** *eLife* **6** [Google Scholar](#)
75. Braumann F, Klug D, Kehrer J, Song G, Feng J, Springer TA, et al. (2023) **Conformational change of Plasmodium TRAP is essential for sporozoite migration and transmission** *EMBO Rep* **24** <https://www.embopress.org/doi/10.15252/embr.202357064> | [Google Scholar](#)
76. Douglas RG, Reinig M, Neale M, Frischknecht F (2018) **Screening for potential prophylactics targeting sporozoite motility through the skin** *Malar J* **17** [PubMed](#) | [Google Scholar](#)
77. Meis JFGM, Verhave JP, Jap PHK, Meuwissen JHET (1985) **Transformation of sporozoites of Plasmodium berghei into exoerythrocytic forms in the liver of its mammalian host** *Cell Tissue Res* **241**:353–60 <https://link.springer.com/article/10.1007/BF00217180> | [Google Scholar](#)
78. Short B (2010) **Polyglutamylation makes the cut** *J Cell Biol* **189**:920 [Google Scholar](#)
79. Genova M, Grycova L, Puttrich V, Magiera MM, Lansky Z, Janke C, et al. (2023) **Tubulin polyglutamylation differentially regulates microtubule-interacting proteins** *EMBO J* **42**:e112101 [Google Scholar](#)
80. Pathak NH, Drummond IA (2009) **Polyglutamylation and the fleer gene** *Methods Cell Biol* **94**:317–32 [Google Scholar](#)
81. Jayabalasingham B, Bano N, Coppens I (2010) **Metamorphosis of the malaria parasite in the liver is associated with organelle clearance** *Cell Res* **20**:1043 [Google Scholar](#)
82. Graewe S, Stanway RR, Rennenberg A, Heussler VT (2012) **Chronicle of a death foretold: Plasmodium liver stage parasites decide on the fate of the host cell** *FEMS Microbiol Rev* **36**:111–30 <https://doi.org/10.1111/j.1574-6976.2011.00297.x> | [Google Scholar](#)
83. de Niz M, Caldelari R, Kaiser G, Zuber B, Heo W Do, Heussler VT, et al. (2021) **Hijacking of the host cell Golgi by Plasmodium berghei liver stage parasites** *J Cell Sci* **134**:jcs252213 [Google Scholar](#)
84. O’neal AJ, Butler LR, Rolandelli A, Gilk SD, Pedra JHF (2020) **Lipid hijacking: A unifying theme in vector-borne diseases** *eLife* **9**:1–31 [Google Scholar](#)
85. Fraser M, Curtis B, Phillips P, Yates PA, Lam KS, Netzel O, et al. (2024) **Harnessing cholesterol uptake of malaria parasites for therapeutic applications** *EMBO Mol Med* **16**:1515–32 <https://www.embopress.org/doi/10.1038/s44321-024-00087-1> | [Google Scholar](#)
86. Schmuckli-Maurer J, Bindschedler AF, Wacker R, Würigler OM, Rehmann R, Lehmborg T, et al. (2024) **Plasmodium berghei liver stage parasites exploit host GABARAP proteins for TFEB**

activation *Communications Biology* **7**:1–13 <https://www.nature.com/articles/s42003-024-07242-x> | [Google Scholar](#)

87. Zeeshan M, Rea E, Abel S, Vukušić K, Markus R, Brady D, et al. (2023) **Plasmodium ARK2 and EB1 drive unconventional spindle dynamics, during chromosome segregation in sexual transmission stages** *Nature Communications* **14**:1–19 <https://www.nature.com/articles/s41467-023-41395-3> | [Google Scholar](#)
88. Van Dijk J, Miro J, Strub JM, Lacroix B, Van Dorsselaer A, Edde B, et al. (2008) **Polyglutamylation Is a Post-translational Modification with a Broad Range of Substrates** *Journal of Biological Chemistry* **283**:3915–22 [Google Scholar](#)
89. Morrisette N, Abbaali I, Ramakrishnan C, Hehl AB (2023) **The Tubulin Superfamily in Apicomplexan Parasites** *Microorganisms* **11** [PubMed](#) | [Google Scholar](#)
90. Kappe SHI, Kaiser K, Matuschewski K (2003) **The Plasmodium sporozoite journey: a rite of passage** *Trends Parasitol* **19**:135–43 [Google Scholar](#)
91. Ferreira JL, Heincke D, Wichers JS, Liffner B, Wilson DW, Gilberger TW (2021) **The Dynamic Roles of the Inner Membrane Complex in the Multiple Stages of the Malaria Parasite** *Front Cell Infect Microbiol* **10**:611801 [Google Scholar](#)
92. Harding CR, Meissner M (2014) **The inner membrane complex through development of Toxoplasma gondii and Plasmodium** *Cell Microbiol* **16**:632 [Google Scholar](#)
93. Burda PC, Roelli MA, Schaffner M, Khan SM, Janse CJ, Heussler VT (2015) **A Plasmodium Phospholipase Is Involved in Disruption of the Liver Stage Parasitophorous Vacuole Membrane** *PLoS Pathog* **11**:e1004760 [Google Scholar](#)
94. Kaiser G, De Niz M, Zuber B, Burda PC, Kornmann B, Heussler VT, et al. (2016) **High resolution microscopy reveals an unusual architecture of the Plasmodium berghei endoplasmic reticulum** *Mol Microbiol* **102**:775–91 [PubMed](#) | [Google Scholar](#)
95. Philip N, Waters AP (2015) **Conditional degradation of plasmodium calcineurin reveals functions in parasite colonization of both host and vector** *Cell Host Microbe* **18**:122–31 <https://www.cell.com/action/showFullText?pii=S1931312815002504> | [Google Scholar](#)
96. Lin J wen, Annoura T, Sajid M, Chevalley-Maurel S, Ramesar J, Klop O, et al. (2011) **A novel “Gene Insertion/Marker Out” (GIMO) method for transgene expression and gene complementation in rodent malaria parasites** *PLoS One* **6** [Google Scholar](#)
97. Janse CJ, Ramesar J, Waters AP (2006) **High-efficiency transfection and drug selection of genetically transformed blood stages of the rodent malaria parasite Plasmodium berghei** *Nature Protocols* **1**:346–56 <https://www.nature.com/articles/nprot.2006.53> | [Google Scholar](#)
98. De Niz M, Stanway RR, Wacker R, Keller D, Heussler VT (2016) **An ultrasensitive NanoLuc-based luminescence system for monitoring Plasmodium berghei throughout its life cycle** *Malaria Journal* **15**:1–24 <https://malariajournal.biomedcentral.com/articles/10.1186/s12936-016-1291-9> | [Google Scholar](#)

99. Salman AM, Mogollon CM, Lin JW, Van Pul FJA, Janse CJ, Khan SM (2015) **Generation of transgenic rodent malaria parasites expressing human malaria parasite proteins** *Methods in Molecular Biology* **1325**:257–86 [Google Scholar](#)
100. Govindarajalu G, Rizvi Z, Kumar D, Sijwali PS (2019) **Lyse-Reseal Erythrocytes for Transfection of Plasmodium falciparum** *Sci Rep* **9** [Google Scholar](#)
101. Pino P, Caldelari R, Mukherjee B, Vahokoski J, Klages N, Maco B, et al. (2017) **A multistage antimalarial targets the plasmepsins IX and X essential for invasion and egress** *Science* (1979) **358**:522–8 <https://www.science.org/doi/10.1126/science.aaf8675> | [Google Scholar](#)
102. Kaur I, Vasudevan A, Rawal P, Tripathi DM, Ramakrishna S, Kaur S, et al. (2023) **Primary Hepatocyte Isolation and Cultures: Technical Aspects, Challenges and Advancements** *Bioengineering* **10**:131 [Google Scholar](#)
103. Bindschedler A, Schmuckli-Maurer J, Buchser S, Fischer TD, Wacker R, Davalan T, et al. (2024) **LC3B labeling of the parasitophorous vacuole membrane of Plasmodium berghei liver stage parasites depends on the V-ATPase and ATG16L1** *Mol Microbiol* **121**:1095–111 <https://onlinelibrary.wiley.com/doi/full/10.1111/mmi.15259> | [Google Scholar](#)
104. Blight J, Sala KA, Atcheson E, Kramer H, El Turabi A, Real E, et al. (2021) **Dissection-independent production of Plasmodium sporozoites from whole mosquitoes** *Life Sci Alliance* **4**:e202101094 [Google Scholar](#)
105. Liffner B, Silva TLA e, Vega-Rodriguez J, Absalon S. (2024) **Mosquito Tissue Ultrastructure-Expansion Microscopy (MoTissU-ExM) enables ultrastructural and anatomical analysis of malaria parasites and their mosquito** *BMC Methods* **1**:1–15 <https://bmcmethods.biomedcentral.com/articles/10.1186/s44330-024-00013-4> | [Google Scholar](#)
106. Rankin KE, Graewe S, Heussler VT, Stanway RR (2010) **Imaging liver-stage malaria parasites** *Cell Microbiol* **12**:569–79 [PubMed](#) | [Google Scholar](#)

Author information

Kodzo Atchou

Institute of Cell Biology, University of Bern, Bern, Switzerland, Graduate School for Cellular and Biomedical Sciences, University of Bern, Bern, Switzerland
ORCID iD: [0000-0003-3264-6351](https://orcid.org/0000-0003-3264-6351)

For correspondence: kodzo.atchou@unibe.ch

Magali Roques

Institute of Cell Biology, University of Bern, Bern, Switzerland

Ruth Rehmann

Institute of Cell Biology, University of Bern, Bern, Switzerland

Reto Caldelari

Institute of Cell Biology, University of Bern, Bern, Switzerland

Melanie Schmid

Institute of Cell Biology, University of Bern, Bern, Switzerland, Graduate School for Cellular and Biomedical Sciences, University of Bern, Bern, Switzerland

Simone Grossi

Institute of Cell Biology, University of Bern, Bern, Switzerland

Bianca Manuela Berger

Institute of Cell Biology, University of Bern, Bern, Switzerland, Graduate School for Cellular and Biomedical Sciences, University of Bern, Bern, Switzerland

Torsten Ochsenreiter

Institute of Cell Biology, University of Bern, Bern, Switzerland

Friedrich Frischknecht

Department of Parasitology, University of Heidelberg Medical School, Heidelberg, Germany
ORCID iD: [0000-0002-8332-6668](https://orcid.org/0000-0002-8332-6668)

Volker Heussler

Institute of Cell Biology, University of Bern, Bern, Switzerland
ORCID iD: [0000-0001-8028-9825](https://orcid.org/0000-0001-8028-9825)

For correspondence: volker.heussler@unibe.ch

Editors

Reviewing Editor

Dominique Soldati-Favre

University of Geneva, Geneva, Switzerland

Senior Editor

Dominique Soldati-Favre

University of Geneva, Geneva, Switzerland

Reviewer #1 (Public review):

The authors try to investigate how the population of microtubules (LSPMB) that originate from sporozoite subpellicular microtubules (SSPM) and are remodelled during liver-stage development of malaria parasites. These bundles shrink over time and help form structures needed for cell division. The authors have used expansion microscopy, live-cell imaging, genetically engineered mutants, and pharmacological perturbation to study parasite development with liver cells.

A major strength of the manuscript is the live cell imaging and expansion microscopy to study this challenging liver stage of parasite development. It gives important knowledge that PTMs of α -tubulin, such as polyglutamylation and tyrosination/detyrosination, are crucial for microtubule stability. Mutations in α -tubulin reduce the parasite's ability to move and proliferate in the liver cells. The drug oryzalin, which targets microtubules, also blocks parasite development, showing how important dynamic microtubules are at this stage.

The major problem in the manuscript was the way it flows, as the authors keep shifting from the liver stage to the sporogony stages and then back to the liver stages. It was very confusing at times to know what the real focus of the study is, whether sporozoite development or liver

stage development. The flow of the manuscript could be improved. Some of the findings reported here substantiate the previous electron microscopy.

Overall, the study represents an important contribution towards understanding cytoskeletal remodelling during liver stage infection. The study suggests that tubulin modifications are key for the parasite's survival in the liver and could be targets for new malaria treatments. This is also the stage that has been used for vaccine development, so any knowledge of how parasites proliferate in the liver cells will be beneficial towards intervention approaches.

<https://doi.org/10.7554/eLife.108209.1.sa3>

Reviewer #2 (Public review):

Summary:

The authors investigated microtubule distribution and their possible post-translational modifications (PTM) in *Plasmodium berghei* during development of the liver stage, using either hepatocytes or HeLa cells as models. They used conventional immunofluorescence assays and expansion microscopy with various antibodies recognising tubulin and, in the second part of the work, its candidate PTMs, as well as markers of *Plasmodium*, in addition to live imaging with a fluorescent marker for tubulin. In the third part of the study, they generated 3 mutants deprived of either the last four residues or the last 11 residues, or where a candidate polyglutamylation site was substituted by an alanine residue.

Strengths:

In the first part, microtubules are monitored by a combination of two approaches (IFA and live), revealing nicely the evolution of the sporozoite subpellicular microtubules (SSPM, the sporozoite is the developmental stage present in salivary glands of the mosquitoes and that infects hepatocytes) into a different structure termed liver-stage parasite microtubule bundle (LSPMB). The LSPMB shrinks during the course of parasite development and finally disappears while hemi-spindles emerge over time. Contact points between these two structures are observed frequently in live cells and occasionally in fixed cells, suggesting the intriguing possibility that tubulin might be recycled from the LSPMB to contribute to hemi-spindle formation.

In the second part, antibodies recognising (1) the final tyrosine found at the C-terminal tail and (2) a stretch of 3 glutamate residues in a side chain are used to monitor these candidate PTMs. Signals are positive at the SSPM, and while it remains positive for polyglutamylation, it becomes negative for the final tyrosine at the LSPM, while a positive signal emerges at hemi-spindles at later stages of development.

In the last part, the three mutants are fed to mosquitoes, where they show reduced development, the one lacking the alpha-tubulin tail even failing to reach the salivary glands. However, the two other mutants infect HeLa cells normally, whereas sporozoites with the C-terminal tail deletion recovered from the haemolymph did not develop in these cells.

The first part provides convincing evidence that microtubules are extensively remodelled during the infection of hepatocytes and HeLa cells, in agreement with the spectacular *Plasmodium* morphogenetic changes accompanying massive and rapid proliferation. The third part brings further confirmation that the C-terminal tail of alpha-tubulin is essential for multiple stages of parasite development, in agreement with previous work (50). Since it is the region where several post-translational modifications take place in other organisms (detyrosination, polyglutamylation, glycylation), it makes sense to propose that the essential function is related to these PTMs also in *Plasmodium*.

Weaknesses:

The significance of tubulin PTM relies on two antibodies whose reactivity to *Plasmodium* tubulins is unclear (see below). The interpretation of the literature on de/tyrosination and polyglutamylation is confusing in several places, meaning that the statements about the possible role of these PTMs need to be carefully revisited.

The authors use the term "tyrosination" but the alpha1-tubulin studied here possesses the final tyrosine when it is synthesised, so it is "tyrosinated" by default. It could potentially be removed by a tyrosine carboxypeptidase of the vasoinhibin family (VASH) as reported in other species. After removal, this tyrosine can be added again by a tubulin-tyrosine ligase (TTL) enzyme. It is therefore more appropriate to talk about de/tyrosination-re/tyrosination rather than tyrosination (this confusion is unfortunately common in the literature, see Janke & Magiera, 2020).

The difficulty here is that there is so far no evidence that de/tyrosination takes place in *Plasmodium*. Neither VASH nor TTL could be identified in the *Plasmodium* genome (ref 31, something we can confirm with our unsuccessful BLAST analyses), and mass spectrometry studies of purified tubulin, albeit from blood stages, did not find evidence for de/tyrosination (reference 43). Western blots using an antibody against de/tyrosinated tubulin did not produce a positive signal, neither on purified tubulin, nor on whole parasites (43). Of course, the situation could be different in liver stages, but the question of the de/tyrosinating enzyme is still there. The existence of a unique *Plasmodium* system for de/tyrosination cannot be formally ruled out, but given the high degree of conservation of these PTMs and their associated enzymes, it sounds difficult to imagine.

The fact that the anti-tyrosinated antibody still produced a signal in the cell line where the final tyrosine is deleted raises issues about its specificity. A cross-reactivity with beta-tubulin is proposed, but the *Plasmodium* beta-tubulin does not carry a final tyrosine, further raising concerns about antibody specificity.

The interpretation of these results should therefore be considered carefully. There also seems to be some confusion in the function of de/tyrosination cited from the literature. It is said in line 229 that "tyrosination has been associated with stable microtubules" (33, 34, 50, 55). References 33 and 34 actually show that tyrosinated microtubules turn over faster in neurons or in epithelial cells, respectively, while references 50 and 55 do not study de/re/tyrosination. The general consensus is that tyrosinated microtubules are more dynamic (see reference 24).

The situation is a bit different for polyglutamylation since several candidate poly- or mono-glutamylases have been identified in the *Plasmodium* genome, and at least mono-glutamylation of beta-tubulin has been formally proven, still in bloodstream stages (ref 43). The authors propose that the residue E445 is the polyglutamylation site. To our knowledge, this has not been demonstrated for *Plasmodium*. This residue is indeed the favourite one in several organisms such as humans and trypanosomes (Eddé et al., Science 1990; Schneider et al., JCS, 1997), and it is tempting to propose it would be the same here. However, TTLs bind the tubulin tails from their C-terminal end like a glove on a finger (Garnham et al., Cell, 2015), and the presence of two extra residues in *Plasmodium* tubulins would mean that the reactive glutamate might be in position E447 rather than E445. This is worth discussing. On the positive side, it is encouraging to see that signals for both anti-tyrosinated tail and poly-glutamylated side chain are going down in the various mutants, but this would need validation with a comparison for alpha-tubulin signal.

Line 316: polyglutamylation "is commonly associated with dynamic microtubule behavior (78-80)". Actually, references 78 and 79 show the impact of this PTM on interaction with spastin, and reference 80 discusses polyglutamylation as a marker of stable microtubules in

the context of cilia and flagella. The consensus is that polyglutamylated microtubules tend to be more stable (ref24).

Conclusion:

The first and the third parts of this manuscript - evolution of microtubules and importance of the C-terminal tails for Plasmodium development - are convincing and well supported by data. However, the presence and role of tubulin PTM should be carefully reconsidered.

Plasmodium tubulins are more closely related to plant tubulins and are sensitive to inhibitors that do not affect mammalian microtubules. They therefore represent promising drug targets as several well-characterised compounds used as herbicides are available. The work produced here further defines the evolution of the microtubule network in sporozoites and liver stages, which are the initial and essential first steps of the infection. Moreover, Plasmodium has multiple specificities that make it a fascinating organism to study both for cell biology and evolution. The data reported here are elegant and will attract the attention of the community working on parasites but also on the cytoskeleton at large. It will be interesting to have the feedback of other people working on tubulin PTMs to figure out the significance of this part of the work.

<https://doi.org/10.7554/eLife.108209.1.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Atchou et al. investigates the role of the microtubule cytoskeleton in sporozoites of Plasmodium berghei, including possible functions of microtubule post-translational modifications (tyrosination and polyglutamylation) in the development of sporozoites in the liver. They also assessed the development of sporozoites in the mosquito. Using cell culture models and in vivo infections with parasites that contain tubulin mutants deficient in certain PTMs, they show that many aspects of the life cycle progression are impaired. The main conclusion is that microtubule PTMs play a major role in the differentiation processes of the parasites.

However, there are a number of major and minor points of criticism that relate to the interpretation of some of the data.

Comments:

(1) The first paragraph of "Results" almost suggests that the presence of a subpellicular MT-array in sporozoites is a new discovery. This is not the case, see e.g. the recent publication by Ferreira et al. (Nature Communications, 2023).

(2) Why were HeLa cells and not hepatocytes (as in Figure 3) used for measuring infection rates of the mutants in Figure 5H and 5L? As I understand, HeLa cells are not natural host cells for invading sporozoites. HeLa cells are epithelial cells derived from a cervical tumour. I am not an expert in Plasmodium biology, but is a HeLa infection an accepted surrogate model for liver stage development?

(3) The tubulin staining in Figures 1A and 1B is confusing and doesn't seem to make sense. Whereas in 1A the antibody nicely stains host and parasite tubulin, in 1B, only parasite tubulin is visible. If the same antibody and the same host cells have been used, HeLa cytoplasmic microtubules should be visible in 1B. In fact, they should be the predominant antigen. The same applies to Figure 2, where host microtubules are also not visible.

(4) In Figures 2A and B, the host nuclei appear to have very different sizes in the DMSO controls and in the drug-treated cells. For example, in the 20 μ M (-) image (bottom right), the nuclei are much larger than in the DMSO (-) control (top left). If this is the case, expansion microscopy hasn't worked reproducibly, and therefore, quantification of fluorescence is problematic. The scalebar is the same for all panels.

(5) I don't quite follow the argument that spindles and the LSPMB are dynamic structures (e.g., lines 145, 174). That is a trivial statement for the spindle, as it is always dynamic, but beyond that, it has only been shown that the structure is sensitive to oryzalin. That says little about any "natural" dynamic behaviour. Any microtubule structure can be destroyed by a particular physical or chemical treatment, but that doesn't mean all structures are dynamic. It also depends on the definition of "dynamic" in a particular context, for example, the time scale of dynamic behaviour (changes within seconds, minutes, or hours).

(6) I am not sure what part in the story EB1 plays. The data are only shown in the Supplements and don't seem to be of particular relevance. EB1 is a ubiquitous protein associated with microtubule plus ends. The statement (line 192) that it "may play a broader role..." is unsubstantiated and cannot be based merely on the observation that it is expressed in a particular life cycle stage.

(7) Line 196 onwards: The antibody IN105 is better known in the field as polyE. Maybe that should be added in Materials and Methods. Also, the antibody T9028 against tyrosinated tubulin is poorly validated in the literature and rarely used. Usually, researchers in this field use the monoclonal antibody YL1/2. I am not sure why this unusual antibody was chosen in this study. In fact, has its specificity against tyrosinated α -tubulin from *Plasmodium berghei* ever been shown? The original antigen was human and had the sequence EGEEY. The *Plasmodium* sequence is YEADY and hence very different. It is stated that the LSPMB is both polyglutamylated and tyrosinated. This is unusual because polyglutamylated microtubules are usually indicative of stable microtubules, whereas tyrosinated microtubules are found on freshly polymerised and dynamic microtubules. However, a co-localisation within the same cell has not been attempted. This is, however, possible since polyE is a rabbit antibody and T9028 is a mouse antibody. I suspect that differences or gradients along the LSPMB would have been noticed. Also, in lines 207/208, it is said that tyrosination disappears after hepatocyte invasion, which is shown in Figure 3. However, in Figure 3A, quite a lot of positive signals for tyrosination are visible in the 54 and 56 hpi panels.

(8) In line 229, it is stated that tyrosination "has previously been associated with stable microtubule in motility". This statement is not correct. In fact, none of the cited references that apparently support this statement show that this is the case. On the contrary, stable microtubules, such as flagellar axonemes, are almost completely detyrosinated. Therefore, tyrosination is a marker for dynamic microtubules, whereas detyrosinated microtubules are indicative of stable microtubules. This is an established fact, and it is odd that the authors claim the opposite.

(9) Line 236 onwards: Concerning the generation of tubulin mutants, I think it is necessary to demonstrate successful replacement of the wild-type allele by the mutant allele. I am sure the authors have done this by amplification and subsequent sequencing of the genomic locus using PCR primers outside the plasmid sequences. I suggest including this information, e.g., by displaying the chromatograph trace in a supplementary figure. Or are the sequences displayed in Figure S3B already derived from sequenced genomic DNA? This is not described in the Legend or in Materials and Methods. The left PCR products obtained for Figure S3 B would be a suitable template for sequencing.

(10) It is also important to be aware of the fact that glutamylation also occurs on β -tubulin. This signal will also be detected by polyE (IN105). Therefore, it is surprising that IN105

immunofluorescence is negative on the C-term Δ cells (Figure S3 D). Is there anything known about confirmed polyglutamylation sites on both α - and β -tubulins in *Plasmodium*, e.g., by MS? In *Toxoplasma*, both α - and β -tubulin have been shown to be polyglutamylated.

(11) Figure S3 is very confusing. In the legend, certain intron deletions are mentioned. How does this relate to posttranslational tubulin modifications? The corresponding section in Results (lines 288-292) is also not very helpful in understanding this.

(12) Figure 4E doesn't look like brightfield microscopy but like some sort of fluorescent imaging. In Figure 4C, were the control (No Δ) cells with an integrated cassette, but no mutations, or non-transgenic cells?

(13) It is difficult to understand why the Ty Δ and the Ct Δ mutants still show quite a strong signal using the anti-tyrosination antibody. If the mutants have replaced all wild-type alleles, the signal should be completely absent, unless the antibody (see my comment above concerning T9028) cross-reacts with detyrosinated microtubules. Therefore, the quantitation in Figures 5F and 5G is actually indicative of something that shouldn't be like that. The quantitation of 5F is at odds with the microscopy image in 5D. If this image is representative, the anti-Ty staining in Ty Δ is as strong as in the control No Δ .

(14) The statement that the failure of Ct Δ mutants to generate viable sporozoites is due to the lack of microtubule PTMs (lines 295-296) is speculative. The lack of the entire C-terminal tail could have a number of consequences, such as impaired microtubule assembly or failure to recruit and bind associated proteins. This is not necessarily linked to PTMs. Also, it has been shown in yeast that for microtubules to form properly and exquisite regulation (proteostasis) of the ratio between α - and β -tubulin is essential (Wethekam and Moore, 2023). I am not sure, but according to Materials and Methods (line 423), the gene cassettes for replacing the wild-type tubulin gene with the mutant versions contain a selectable marker gene for pyrimethamine selection. Are there qPCR data that show that expression levels of mutant α -tubulin are more or less the same as the wild-type levels?

(15) In the Discussion, my impression is that two recent studies, the superb Expansion Microscopy study by Bertiaux et al. (2021) and the cryo-EM study by Ferreira et al. (2023), are not sufficiently recognised (although they are cited elsewhere in the manuscript). The latter study includes a detailed description of the microtubule cytoskeleton in sporozoites. However, the present study clearly expands the knowledge about the structure of the cytoskeleton in liver stage parasites and is one of the few studies addressing the distribution and function of microtubule post-translational modifications in *Plasmodium*.

(16) I somewhat disagree with the statement of a co-occurrence of polyglutamylated and tyrosinated microtubules. I think the resolution is too low to reach that conclusion. As this is a bold claim, and would be contrary to what is known from other organisms, it would require a more rigorous validation. Given the apparent problems with the anti-Ty antibody (signal in the Ty Δ mutant), one should be very cautious with this claim.

(17) In the Discussion (lines 311 and 377), it is again claimed that tyrosinated microtubules are "a well-known marker of stable microtubules". This statement is completely incorrect, and I am surprised by this serious mistake. A few lines later, the authors say that polyglutamylated is "commonly associated with dynamic microtubule behaviour". Again, this is completely incorrect and is the opposite of what is firmly established in the literature. Polyglutamylation and detyrosination are markers of stable microtubules.

(18) In line 339, the authors interpret the residual antibody staining after the introduction of the mutant tubulin as a compensatory mechanism. There is no evidence for this. More likely explanations are firstly the quality of the anti-Ty-antibody used (see comment above), and the fact that also β -tubulin carries C-terminal polyglutamylation sites, which haven't been

investigated in this study. PTMs on β -tubulin are not compensatory, but normal PTMs, at least in all other organisms where microtubule PTMs have been investigated.

<https://doi.org/10.7554/eLife.108209.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

The authors try to investigate how the population of microtubules (LSPMB) that originate from sporozoite subpellicular microtubules (SSPM) and are remodelled during liver-stage development of malaria parasites. These bundles shrink over time and help form structures needed for cell division. The authors have used expansion microscopy, live-cell imaging, genetically engineered mutants, and pharmacological perturbation to study parasite development with liver cells.

A major strength of the manuscript is the live cell imaging and expansion microscopy to study this challenging liver stage of parasite development. It gives important knowledge that PTMs of α -tubulin, such as polyglutamylation and tyrosination/detyrosination, are crucial for microtubule stability. Mutations in α -tubulin reduce the parasite's ability to move and proliferate in the liver cells. The drug oryzalin, which targets microtubules, also blocks parasite development, showing how important dynamic microtubules are at this stage.

The major problem in the manuscript was the way it flows, as the authors keep shifting from the liver stage to the sporogony stages and then back to the liver stages. It was very confusing at times to know what the real focus of the study is, whether sporozoite development or liver stage development. The flow of the manuscript could be improved. Some of the findings reported here substantiate the previous electron microscopy.

Overall, the study represents an important contribution towards understanding cytoskeletal remodelling during liver stage infection. The study suggests that tubulin modifications are key for the parasite's survival in the liver and could be targets for new malaria treatments. This is also the stage that has been used for vaccine development, so any knowledge of how parasites proliferate in the liver cells will be beneficial towards intervention approaches.

We would like to express our sincere gratitude to Reviewer #1 for the positive and encouraging feedback on our manuscript. We are delighted that the reviewer found our experimental design and methodologies appropriate and that our study represents an important contribution to understanding cytoskeletal remodelling during liver stage infection, a critical phase for vaccine development. We are also grateful to the reviewer for highlighting the issue with the manuscript's flow. We acknowledge this limitation and will significantly improve the narrative structure and logical progression in the revised manuscript to ensure clarity and avoid any potential confusion. Thank you again for your thoughtful and constructive comments.

Reviewer #2 (Public review):

Summary:

*The authors investigated microtubule distribution and their possible post-translational modifications (PTM) in *Plasmodium berghei* during development of the liver stage, using*

either hepatocytes or HeLa cells as models. They used conventional immunofluorescence assays and expansion microscopy with various antibodies recognising tubulin and, in the second part of the work, its candidate PTMs, as well as markers of *Plasmodium*, in addition to live imaging with a fluorescent marker for tubulin. In the third part of the study, they generated 3 mutants deprived of either the last four residues or the last 11 residues, or where a candidate polyglutamylation site was substituted by an alanine residue.

Strengths:

In the first part, microtubules are monitored by a combination of two approaches (IFA and live), revealing nicely the evolution of the sporozoite subpellicular microtubules (SSPM, the sporozoite is the developmental stage present in salivary glands of the mosquitoes and that infects hepatocytes) into a different structure termed liver-stage parasite microtubule bundle (LSPMB). The LSPMB shrinks during the course of parasite development and finally disappears while hemi-spindles emerge over time. Contact points between these two structures are observed frequently in live cells and occasionally in fixed cells, suggesting the intriguing possibility that tubulin might be recycled from the LSPMB to contribute to hemi-spindle formation.

In the second part, antibodies recognising (1) the final tyrosine found at the C-terminal tail and (2) a stretch of 3 glutamate residues in a side chain are used to monitor these candidate PTMs. Signals are positive at the SSPM, and while it remains positive for polyglutamylation, it becomes negative for the final tyrosine at the LSPM, while a positive signal emerges at hemi-spindles at later stages of development.

In the last part, the three mutants are fed to mosquitoes, where they show reduced development, the one lacking the alpha-tubulin tail even failing to reach the salivary glands. However, the two other mutants infect HeLa cells normally, whereas sporozoites with the C-terminal tail deletion recovered from the haemolymph did not develop in these cells.

*The first part provides convincing evidence that microtubules are extensively remodelled during the infection of hepatocytes and HeLa cells, in agreement with the spectacular *Plasmodium* morphogenetic changes accompanying massive and rapid proliferation. The third part brings further confirmation that the C-terminal tail of alpha-tubulin is essential for multiple stages of parasite development, in agreement with previous work (50). Since it is the region where several post-translational modifications take place in other organisms (detyrosination, polyglutamylation, glycylation), it makes sense to propose that the essential function is related to these PTMs also in *Plasmodium*.*

Weaknesses:

*The significance of tubulin PTM relies on two antibodies whose reactivity to *Plasmodium* tubulins is unclear (see below). The interpretation of the literature on detyrosination and polyglutamylation is confusing in several places, meaning that the statements about the possible role of these PTMs need to be carefully revisited.*

The authors use the term "tyrosination" but the alpha1-tubulin studied here possesses the final tyrosine when it is synthesised, so it is "tyrosinated" by default. It could potentially be removed by a tyrosine carboxypeptidase of the vasoinhibin family (VASH) as reported in other species. After removal, this tyrosine can be added again by a tubulin-tyrosine ligase (TTL) enzyme. It is therefore more appropriate to talk about detyrosination-retyrosination rather than tyrosination (this confusion is unfortunately common in the literature, see Janke & Magiera, 2020).

The difficulty here is that there is so far no evidence that detyrosination takes place in Plasmodium. Neither VASH nor TTL could be identified in the Plasmodium genome (ref 31, something we can confirm with our unsuccessful BLAST analyses), and mass spectrometry studies of purified tubulin, albeit from blood stages, did not find evidence for detyrosination (reference 43). Western blots using an antibody against detyrosinated tubulin did not produce a positive signal, neither on purified tubulin, nor on whole parasites (43). Of course, the situation could be different in liver stages, but the question of the detyrosinating enzyme is still there. The existence of a unique Plasmodium system for detyrosination cannot be formally ruled out but given the high degree of conservation of these PTMs and their associated enzymes, it sounds difficult to imagine.

The fact that the anti-tyrosinated antibody still produced a signal in the cell line where the final tyrosine is deleted raises issues about its specificity. A cross-reactivity with beta-tubulin is proposed, but the Plasmodium beta-tubulin does not carry a final tyrosine, further raising concerns about antibody specificity.

The interpretation of these results should therefore be considered carefully. There also seems to be some confusion in the function of detyrosination cited from the literature. It is said in line 229 that "tyrosination has been associated with stable microtubules" (33, 34, 50, 55). References 33 and 34 actually show that tyrosinated microtubules turn over faster in neurons or in epithelial cells, respectively, while references 50 and 55 do not study de/retyrosination. The general consensus is that tyrosinated microtubules are more dynamic (see reference 24).

The situation is a bit different for polyglutamylation since several candidate poly- or mono-glutamylases have been identified in the Plasmodium genome, and at least mono-glutamylation of beta-tubulin has been formally proven, still in bloodstream stages (ref 43). The authors propose that the residue E445 is the polyglutamylation site. To our knowledge, this has not been demonstrated for Plasmodium. This residue is indeed the favourite one in several organisms such as humans and trypanosomes (Eddé et al., Science 1990; Schneider et al., JCS, 1997), and it is tempting to propose it would be the same here. However, TTLs bind the tubulin tails from their C-terminal end like a glove on a finger (Garnham et al., Cell, 2015), and the presence of two extra residues in Plasmodium tubulins would mean that the reactive glutamate might be in position E447 rather than E445. This is worth discussing.

On the positive side, it is encouraging to see that signals for both anti-tyrosinated tail and poly-glutamylated side chain are going down in the various mutants, but this would need validation with a comparison for alpha-tubulin signal.

Line 316: polyglutamylation "is commonly associated with dynamic microtubule behavior (78-80)". Actually, references 78 and 79 show the impact of this PTM on interaction with spastin, and reference 80 discusses polyglutamylation as a marker of stable microtubules in the context of cilia and flagella. The consensus is that polyglutamylated microtubules tend to be more stable (ref24).

Conclusion:

The first and the third parts of this manuscript - evolution of microtubules and importance of the C-terminal tails for Plasmodium development - are convincing and well supported by data. However, the presence and role of tubulin PTM should be carefully reconsidered.

Plasmodium tubulins are more closely related to plant tubulins and are sensitive to inhibitors that do not affect mammalian microtubules. They therefore represent promising drug targets as several well-characterised compounds used as herbicides are

available. The work produced here further defines the evolution of the microtubule network in sporozoites and liver stages, which are the initial and essential first steps of the infection. Moreover, Plasmodium has multiple specificities that make it a fascinating organism to study both for cell biology and evolution. The data reported here are elegant and will attract the attention of the community working on parasites but also on the cytoskeleton at large. It will be interesting to have the feedback of other people working on tubulin PTMs to figure out the significance of this part of the work.

We thank Reviewer #2 for the thoughtful and detailed evaluation of our manuscript. We are pleased that the reviewer found our study elegant and believe it will attract the attention of the broader scientific community, both those working on parasites and those focused on cytoskeleton biology. We also acknowledge the concerns raised regarding the specificity of the antibodies used to detect tubulin post-translational modifications (PTMs), as well as the interpretation of their signals and the current lack of identified detyrosination enzymes in the Plasmodium genome. We agree that these are important limitations, and we will address them thoroughly in the revised manuscript. This includes clarifying our interpretation of tyrosination versus detyrosination, adjusting our claims regarding polyglutamylation sites, and carefully revisiting the literature cited to ensure accurate contextualization of PTM function in microtubule stability.

We are grateful for the reviewer's close reading and critical feedback, which will help us substantially improve the clarity, precision, and strength of our manuscript.

Reviewer #3 (Public review):

Summary:

The manuscript by Atchou et al. investigates the role of the microtubule cytoskeleton in sporozoites of Plasmodium berghei, including possible functions of microtubule post-translational modifications (tyrosination and polyglutamylation) in the development of sporozoites in the liver. They also assessed the development of sporozoites in the mosquito. Using cell culture models and in vivo infections with parasites that contain tubulin mutants deficient in certain PTMs, they show that many aspects of the life cycle progression are impaired. The main conclusion is that microtubule PTMs play a major role in the differentiation processes of the parasites.

However, there are a number of major and minor points of criticism that relate to the interpretation of some of the data.

We thank Reviewer #3 for the overall positive assessment of our study and for recognizing its contribution to advancing our understanding of Plasmodium biology and malaria pathogenesis. We appreciate the reviewer's constructive feedback, particularly regarding the interpretation of some of our data. These comments have been very helpful in guiding our revisions, and we have worked to improve both the clarity of our presentation and the precision of our interpretations in the revised manuscript.

Below, we respond in detail to each of the reviewer's points.

Comments:

(1) The first paragraph of "Results" almost suggests that the presence of a subpellicular MT-array in sporozoites is a new discovery. This is not the case, see e.g. the recent publication by Ferreira et al. (Nature Communications, 2023).

We thank the reviewer for pointing this out and fully agree that the subpellicular microtubule (SPM) array in sporozoites is well established, as documented in earlier work (e.g., Cyrklaff et al., 2007) and more recently by Ferreira et al. (Nat. Commun., 2023). Our

intention was not to suggest that the existence of the SSPM is a novel finding. Rather, our study builds on this existing knowledge by demonstrating that these sporozoite-derived microtubules are not disassembled upon hepatocyte entry but are repurposed into a newly described structure, the liver stage parasite microtubule bundle (LSPMB). This reorganization, its persistence into liver stage development, and its dynamic role in microtubule remodeling and nuclear division are, to our knowledge, novel observations. We will revise the manuscript to make this distinction clearer in the introduction and the results section.

(2) Why were HeLa cells and not hepatocytes (as in Figure 3) used for measuring infection rates of the mutants in Figure 5H and 5L? As I understand, HeLa cells are not natural host cells for invading sporozoites. HeLa cells are epithelial cells derived from a cervical tumour. I am not an expert in Plasmodium biology, but is a HeLa infection an accepted surrogate model for liver stage development?

We appreciate the opportunity to clarify our experimental model. While HeLa cells are not the natural host cells, they are a well-established and validated in vitro model for studying *Plasmodium berghei* liver stage development in our lab and others. In this system, the parasite completes its full development and generates infectious merozoites. Numerous studies have successfully used HeLa cells as a liver stage infection model, with key findings subsequently validated in primary hepatocytes or in vivo, confirming its utility as a representative model. We employed this cell line primarily to reduce animal usage in accordance with the 3Rs principles (Replacement, Reduction, Refinement). Importantly, to ensure the biological relevance of our discoveries in HeLa cells, we validated our key findings in primary mouse hepatocytes, as shown in Figure 3. Furthermore, we confirmed the in vivo infectivity of mutant parasite lines that produced typical salivary gland sporozoites through an in vivo infection assay, presented in Figure S4C.

(3) The tubulin staining in Figures 1A and 1B is confusing and doesn't seem to make sense. Whereas in 1A the antibody nicely stains host and parasite tubulin, in 1B, only parasite tubulin is visible. If the same antibody and the same host cells have been used, HeLa cytoplasmic microtubules should be visible in 1B. In fact, they should be the predominant antigen. The same applies to Figure 2, where host microtubules are also not visible.

We thank the reviewer for this careful observation regarding the α -tubulin staining in Figures 1A and 1B. The same host cell type (HeLa) and α -tubulin antibody were indeed used in both experiments. Figure 1A shows results from conventional immunofluorescence assays, where both host and parasite microtubules are clearly stained. In contrast, Figure 1B shows the outcome of ultrastructure expansion microscopy (U-ExM), where parasite microtubules appear prominently, while host microtubules are less visible.

This effect appears to be a technical outcome of the U-ExM protocol, which can differentially preserve or reveal microtubule epitopes. We consistently observed stronger parasite signal across various cell types, including primary hepatocytes (Figure 3A,B). The lack of visible host microtubules in some U-ExM images does not reflect their absence, but rather reduced signal intensity relative to the parasite structures. This is not observed with all antibodies, e.g., host microtubules stain strongly with anti-tyrosinated α -tubulin (Figure 3B), likely reflecting their high tyrosination state.

To overcome this limitation, we employed PS-ExM and combined PS-ExM/U-ExM approaches (as described in reference 56), which allowed simultaneous high-resolution visualization of both host and parasite microtubule networks. These combined methods are now being used in follow-up studies to investigate host-parasite microtubule interactions in more detail.

We will clarify this point in the revised manuscript to avoid confusion.

(4) In Figures 2A and B, the host nuclei appear to have very different sizes in the DMSO controls and in the drug-treated cells. For example, in the 20 μ M (-) image (bottom right), the nuclei are much larger than in the DMSO (-) control (top left). If this is the case, expansion microscopy hasn't worked reproducibly, and therefore, quantification of fluorescence is problematic. The scalebar is the same for all panels.

The expansion microscopy methods used in this study have been rigorously validated for both reproducibility and isotropicity. However, as the reviewer rightly notes, host cell nuclei can vary in size due to several factors, including cell cycle stage, infection status, and the extent of parasite development, all of which can influence host nuclei morphology and size.

Importantly, the quantifications relevant to our conclusions were focused specifically on parasite structures. We did not rely on host nuclear size or host fluorescence intensity as a quantitative readout in this context. While we acknowledge the observed variability in host nuclear dimensions, it does not compromise the accuracy or reproducibility of the parasite specific measurements central to our study.

We will clarify this point in the revised figure legend and manuscript.

(5) I don't quite follow the argument that spindles and the LSPMB are dynamic structures (e.g., lines 145, 174). That is a trivial statement for the spindle, as it is always dynamic, but beyond that, it has only been shown that the structure is sensitive to oryzalin. That says little about any "natural" dynamic behaviour. Any microtubule structure can be destroyed by a particular physical or chemical treatment, but that doesn't mean all structures are dynamic. It also depends on the definition of "dynamic" in a particular context, for example, the time scale of dynamic behaviour (changes within seconds, minutes, or hours).

We agree that sensitivity to chemical depolymerization alone does not necessarily indicate dynamic behavior, particularly in the absence of data on turnover kinetics or temporal changes.

Our interpretation was based on two observations: first, that the LSPMB, which derives from the highly stable sporozoite subpellicular microtubules (known to be drug-resistant), becomes susceptible to depolymerization during the liver stage; and second, that the LSPMB gradually shrinks over time during parasite development. These features suggested a transition toward a more dynamic state compared to its origin. However, we fully agree that “dynamic” is a context-dependent term and that direct evidence such as turnover rates or structural changes on short time scales, is required to rigorously define microtubule dynamics.

We will revise the manuscript to clarify our use of this term and explicitly acknowledge the need for further studies to characterize the timescale and mechanisms underlying LSPMB remodeling.

(6) I am not sure what part in the story EB1 plays. The data are only shown in the Supplements and don't seem to be of particular relevance. EB1 is a ubiquitous protein associated with microtubule plus ends. The statement (line 192) that it "may play a broader role..." is unsubstantiated and cannot be based merely on the observation that it is expressed in a particular life cycle stage.

We agree that EB1 is a ubiquitous microtubule plus-end binding protein and that its presence alone does not imply a novel function. Previous studies (e.g., Maurer et al., 2023; Yang et al.,

2023; Zeeshan et al., 2023) have focused on its role during Plasmodium sexual stages, while its expression during liver and mosquito stages has not been previously documented.

Our data extend this knowledge by showing that EB1 is also expressed during liver stage development, particularly during the highly mitotic schizont phase. While we agree that this observation alone does not prove functional involvement, it raises the possibility of a broader role for EB1 in regulating microtubule dynamics beyond sexual stages. To avoid overinterpretation, we have presented these findings in the supplementary material and will revise the manuscript to tone down speculative statements and clearly frame this as a preliminary observation that warrants further investigation.

(7) Line 196 onwards: The antibody IN105 is better known in the field as polyE. Maybe that should be added in Materials and Methods. Also, the antibody T9028 against tyrosinated tubulin is poorly validated in the literature and rarely used. Usually, researchers in this field use the monoclonal antibody YL1/2. I am not sure why this unusual antibody was chosen in this study. In fact, has its specificity against tyrosinated α -tubulin from Plasmodium berghei ever been shown? The original antigen was human and had the sequence EGEEY. The Plasmodium sequence is YEADY and hence very different. It is stated that the LSPMB is both polyglutamylated and tyrosinated. This is unusual because polyglutamylated microtubules are usually indicative of stable microtubules, whereas tyrosinated microtubules are found on freshly polymerised and dynamic microtubules. However, a co-localisation within the same cell has not been attempted. This is, however, possible since polyE is a rabbit antibody and T9028 is a mouse antibody. I suspect that differences or gradients along the LSPMB would have been noticed. Also, in lines 207/208, it is said that tyrosination disappears after hepatocyte invasion, which is shown in Figure 3. However, in Figure 3A, quite a lot of positive signals for tyrosination are visible in the 54 and 56 hpi panels.

First, we acknowledge that the IN105 antibody is more widely known as "polyE" in the field. We will update the Materials and Methods section accordingly to reflect this nomenclature.

Regarding the use of the T9028 antibody against tyrosinated α -tubulin: we agree that this monoclonal antibody is less commonly used than YL1/2, and we appreciate the reviewer drawing attention to this. The original antigen for T9028 is based on the mammalian C-terminal sequence EGEEY, which differs from the Plasmodium α 1-tubulin sequence (YEADY). Like many in the field, we face the challenge that most available antibodies are raised against mammalian epitopes, and specificity in Plasmodium can vary. Nonetheless, the literature (e.g., Hirst et al., 2022; Fennell et al., 2008) has demonstrated that tyrosination occurs in Plasmodium α 1-tubulin, using anti-tyrosination antibodies including YL1/2.

Following the reviewer's excellent suggestion, we are currently repeating the key experiments using the YL1/2 antibody to compare staining patterns directly with those obtained using T9028. We will include these results in the revised manuscript.

Concerning the potential co-localization of polyglutamylation and tyrosination on the LSPMB: we agree that this is an interesting and testable hypothesis. In the current manuscript, Figures 3A and 3B were generated from independent experiments, and thus co-localization was not assessed. However, as the reviewer correctly notes, polyE and T9028 antibodies are raised in rabbit and mouse, respectively, making co-staining feasible. We will follow up on this experimentally and, if feasible within our revision timeline, include data in the revised version or highlight this as a future direction.

Finally, with regard to Figure 3 and the observation that tyrosination appears to persist at 54 and 56 hpi (Figure 3B): the reviewer is correct that tyrosination signal is still detectable at these time points. Our statement that tyrosination "disappears after hepatocyte invasion" was intended to refer to an overall decrease in signal intensity during early liver stage

development, with a reappearance at later stages (e.g., cytomere formation). We will rephrase this section for greater clarity and ensure that figure annotations and legends unambiguously reflect the dynamics observed.

(8) In line 229, it is stated that tyrosination "has previously been associated with stable microtubule in motility". This statement is not correct. In fact, none of the cited references that apparently support this statement show that this is the case. On the contrary, stable microtubules, such as flagellar axonemes, are almost completely detyrosinated. Therefore, tyrosination is a marker for dynamic microtubules, whereas detyrosinated microtubules are indicative of stable microtubules. This is an established fact, and it is odd that the authors claim the opposite.

We fully agree that in canonical eukaryotic systems, tyrosinated microtubules are generally markers of dynamic microtubule populations, whereas detyrosinated microtubules are typically associated with stability particularly in structures such as flagellar axonemes.

Our original statement will be corrected. In our study, we observed that tyrosinated microtubules are prevalent in invasive stages (sporozoites and merozoites), while detyrosinated forms become more prominent during intracellular liver stage development. This pattern is consistent with the established link between tyrosination and dynamic microtubules.

What is particularly intriguing in *Plasmodium* is the apparent cycling of tyrosination despite the absence of known tubulin tyrosine ligase (TTL) homologs in the genome. This suggests either a highly divergent enzyme or the involvement of host cell factors, a hypothesis supported by the reappearance of tyrosinated microtubules during liver stage schizogony (Figure 3B).

We will revise the relevant text and the Discussion section to reflect these mechanistic considerations more accurately and to avoid misrepresenting established principles of microtubule biology.

(9) Line 236 onwards: Concerning the generation of tubulin mutants, I think it is necessary to demonstrate successful replacement of the wild-type allele by the mutant allele. I am sure the authors have done this by amplification and subsequent sequencing of the genomic locus using PCR primers outside the plasmid sequences. I suggest including this information, e.g., by displaying the chromatograph trace in a supplementary figure. Or are the sequences displayed in Figure S3B already derived from sequenced genomic DNA? This is not described in the Legend or in Materials and Methods. The left PCR products obtained for Figure S3 B would be a suitable template for sequencing.

Indeed, these data are presented in Figure 4B and the corresponding sequence data are shown in Figure S3B. We appreciate the reviewer's suggestion, which will help improve the transparency and reproducibility of our methodology.

*(10) It is also important to be aware of the fact that glutamylation also occurs on β -tubulin. This signal will also be detected by polyE (IN105). Therefore, it is surprising that IN105 immunofluorescence is negative on the C-term Δ cells (Figure S3 D). Is there anything known about confirmed polyglutamylation sites on both α - and β -tubulins in *Plasmodium*, e.g., by MS? In *Toxoplasma*, both α - and β -tubulin have been shown to be polyglutamylated.*

Indeed, polyglutamylation is known to occur not only on α -tubulin but also on β -tubulin in many organisms, including *Toxoplasma gondii*, and the polyE (IN105) antibody is expected to

detect polyglutamylation on both tubulin isoforms.

The parasites shown in Figure S3D correspond to mutant lines originally generated by Spreng et al. (2019): the Intron Δ mutant (with deletion of introns in the Plasmodium α 1-tubulin gene) and the C-term Δ mutant (with deletion of the final three C-terminal residues: ADY). As the reviewer correctly notes, this particular C-terminal deletion does not include the predicted polyglutamylation site (E445 or E447, depending on alignment), and thus should not abolish all polyglutamylation. However, in our experiments, the IN105 signal is substantially reduced in this mutant. This may suggest that structural alterations in the tubulin tail affect accessibility of the polyglutamylation epitope or influence the modification itself though we cannot exclude other possibilities, including changes in antibody recognition.

To date, polyglutamylation sites in Plasmodium tubulins have not been definitively confirmed by mass spectrometry. However, a recent MS-based study (reference 43) detected monoglutamylation on β -tubulin in blood stage parasites. Direct MS evidence for polyglutamylation of either α - or β -tubulin in Plasmodium liver stages is still lacking. We will clarify these points in the revised manuscript to avoid potential confusion and to highlight the need for future biochemical validation of PTM sites.

(11) Figure S3 is very confusing. In the legend, certain intron deletions are mentioned. How does this relate to posttranslational tubulin modifications? The corresponding section in Results (lines 288-292) is also not very helpful in understanding this.

The parasite lines shown in Figure S3D were originally generated by Spreng et al. (2019) and are not directly part of the main set of PTM-targeted mutants described in our study. Specifically, the Intron Δ line carries deletions in introns of the Plasmodium α 1-tubulin gene, while the C-term Δ line lacks the final three C-terminal residues (ADY). These lines were included for comparative purposes to explore whether structural changes in α -tubulin could impact polyglutamylation signal, as detected by the polyE (IN105) antibody.

We acknowledge that the figure legend and corresponding text (lines 288–292) did not adequately explain the rationale for including these control lines. We will revise both the legend and Results section to more clearly describe the origin, purpose, and relevance of these mutants to the overall study.

(12) Figure 4E doesn't look like brightfield microscopy but like some sort of fluorescent imaging. In Figure 4C, were the control (No Δ) cells with an integrated cassette, but no mutations, or non-transgenic cells?

The reviewer is absolutely correct: Figure 4E shows a fluorescent image acquired using widefield microscopy and not a brightfield image. We will revise the figure legend accordingly to avoid confusion. The “BF” (brightfield) label applies only to the left panel in Figure 4C, which depicts oocysts imaged using transmitted light.

Regarding the controls labeled “No Δ ” in Figure 4C, we confirm that these parasites contain the integrated selection cassette but do not harbor any mutations in the target gene. They serve as proper integration controls, allowing us to distinguish the effects of the point mutations or deletions introduced in the experimental lines.

(13) It is difficult to understand why the Ty Δ and the Ct Δ mutants still show quite a strong signal using the anti-tyrosination antibody. If the mutants have replaced all wild-type alleles, the signal should be completely absent, unless the antibody (see my comment above concerning T9028) cross-reacts with detyrosinated microtubules. Therefore, the quantitation in Figures 5F and 5G is actually indicative of something that shouldn't be

like that. The quantitation of 5F is at odds with the microscopy image in 5D. If this image is representative, the anti-Ty staining in TyΔ is as strong as in the control NoΔ.

We agree that the persistence of anti-tyrosination signal in the TyΔ and CtΔ mutant lines is unexpected, given that all wild-type alleles were replaced. This discrepancy has led us to further investigate the specificity of the T9028 antibody, as raised in the reviewer's earlier comment. To address this concern, we are currently repeating the key experiments using the well-established YL1/2 monoclonal antibody, which is widely accepted for detecting tyrosinated α-tubulin in other systems.

We also acknowledge that Figure 5F shows residual tyrosination signal, and the reviewer is correct that this should not occur if the modified residues are the exclusive PTM sites. One possible explanation is that adjacent residues or even alternative tubulin isoforms may serve as substrates. While α1-tubulin is the dominant isoform in *Plasmodium*, low-level expression of α2-tubulin has been detected in liver stages based on transcriptomic data, and it may contribute to the observed signal.

Regarding the apparent discrepancy between the quantification in Figure 5F and the representative image in Figure 5D, we will revise the figure legend to clarify that image selection aimed to show detectable signal, not necessarily the average phenotype. We will also reassess and, if needed, repeat the quantification with improved image sets to ensure accuracy and consistency.

We will revise the manuscript to reflect these points and include a more nuanced interpretation of the residual staining in the mutant lines.

(14) The statement that the failure of CtΔ mutants to generate viable sporozoites is due to the lack of microtubule PTMs (lines 295-296) is speculative. The lack of the entire C-terminal tail could have a number of consequences, such as impaired microtubule assembly or failure to recruit and bind associated proteins. This is not necessarily linked to PTMs. Also, it has been shown in yeast that for microtubules to form properly and exquisite regulation (proteostasis) of the ratio between α- and β-tubulin is essential (Wethekam and Moore, 2023). I am not sure, but according to Materials and Methods (line 423), the gene cassettes for replacing the wild-type tubulin gene with the mutant versions contain a selectable marker gene for pyrimethamine selection. Are there qPCR data that show that expression levels of mutant α-tubulin are more or less the same as the wild-type levels?

We agree that attributing the developmental failure of the CtΔ mutants solely to the absence of microtubule post-translational modifications (PTMs) is speculative. As the reviewer rightly points out, deletion of the entire C-terminal tail may have multiple effects, including impaired microtubule assembly, altered α/β-tubulin stoichiometry, or disruption of interactions with essential microtubule-associated proteins (MAPs). These consequences may arise independently of PTMs.

That said, we note that PTMs particularly polyglutamylation, can modulate MAP binding by altering the surface charge of microtubules (Genova et al., 2023; Mitchell et al., 2010). Therefore, while PTM loss may be a contributing factor, we acknowledge that the phenotype likely results from a combination of mechanisms. We will revise the relevant section of the manuscript to present a more cautious and balanced interpretation.

Regarding the reviewer's question on expression levels: although the replacement constructs include a pyrimethamine resistance cassette, we have not yet quantified α-tubulin transcript levels by qPCR. In the interim, the study by Spreng et al. (2019) (reference 50) on a related α1-tubulin mutations provides valuable insight. They observed no difference in mRNA levels in day 12 oocysts, yet reported fainter microtubule staining and shorter sporozoites, suggesting

a post-transcriptional mechanism affecting protein expression or function in later stages. Furthermore, the phenotypic spectrum across their mutant panel (Suppl. Fig. 3 D and E) implies that robust α -tubulin regulation is highly sensitive to specific sequences.

We acknowledge this as a current limitation in our study and will address it in the revised manuscript, noting that direct measurement of transcript levels is a key area for future investigation.

(15) In the Discussion, my impression is that two recent studies, the superb Expansion Microscopy study by Bertiaux et al. (2021) and the cryo-EM study by Ferreira et al. (2023), are not sufficiently recognised (although they are cited elsewhere in the manuscript). The latter study includes a detailed description of the microtubule cytoskeleton in sporozoites. However, the present study clearly expands the knowledge about the structure of the cytoskeleton in liver stage parasites and is one of the few studies addressing the distribution and function of microtubule post-translational modifications in Plasmodium.

Indeed, our work builds upon the established knowledge from Bertiaux et al. (2021) and the cryo-EM study by Ferreira et al. (2023), as rightly mentioned by the reviewer. We agree that these foundational studies, combined with our findings, will significantly expand the understanding of Plasmodium biology and cytoskeleton dynamics across its life cycle and will open the door for further investigations. We are grateful for this suggestion and will ensure these key studies are appropriately acknowledged in the revised manuscript.

(16) I somewhat disagree with the statement of a co-occurrence of polyglutamylated and tyrosinated microtubules. I think the resolution is too low to reach that conclusion. As this is a bold claim, and would be contrary to what is known from other organisms, it would require a more rigorous validation. Given the apparent problems with the anti-Ty antibody (signal in the TyΔ mutant), one should be very cautious with this claim.

This is a very important point to clarify. As mentioned previously, the initial experiments for these modifications were performed independently. It is established that sporozoite subpellicular microtubules exhibit both tyrosination and polyglutamylation. We will revise the manuscript to temper this statement and clearly indicate that the co-occurrence of these PTMs remains a hypothesis that requires more rigorous validation. As suggested, we are now conducting additional co-staining experiments using the better validated YL1/2 antibody to re-express and directly compare the distribution of both PTMs within the same cell. These follow-up experiments will help clarify whether both modifications occur simultaneously on the same microtubule structures in Plasmodium liver stages.

(17) In the Discussion (lines 311 and 377), it is again claimed that tyrosinated microtubules are "a well-known marker of stable microtubules". This statement is completely incorrect, and I am surprised by this serious mistake. A few lines later, the authors say that polyglutamylated is "commonly associated with dynamic microtubule behaviour". Again, this is completely incorrect and is the opposite of what is firmly established in the literature. Polyglutamylation and detyrosination are markers of stable microtubules.

Indeed, in canonical eukaryotic systems, tyrosinated microtubules are generally considered markers of dynamic microtubule populations, whereas detyrosinated and polyglutamylated microtubules are more commonly associated with stability.

We acknowledge this mistake and will revise the Discussion to correct these statements accordingly. In the context of Plasmodium, our observations suggest an unusual regulation of microtubule dynamics, which may reflect parasite-specific adaptations. For example, we

observed tyrosinated α -tubulin in the stable subpellicular microtubules of sporozoites structures typically known for their exceptional stability. This atypical association implies either non-canonical roles for tyrosination or parasite-specific mechanisms for modulating microtubule properties. Additionally, the presence of both PTMs at different stages of development and on different microtubule populations suggests tightly regulated spatial and temporal modulation of microtubule function.

We will carefully revise the relevant sections of the manuscript to remove incorrect generalizations and ensure accurate representation of the current consensus in the field, while emphasizing the possibility of Plasmodium-specific adaptations that merit further study.

(18) In line 339, the authors interpret the residual antibody staining after the introduction of the mutant tubulin as a compensatory mechanism. There is no evidence for this. More likely explanations are firstly the quality of the anti-Ty-antibody used (see comment above), and the fact that also β -tubulin carries C-terminal polyglutamylation sites, which haven't been investigated in this study. PTMs on β -tubulin are not compensatory, but normal PTMs, at least in all other organisms where microtubule PTMs have been investigated.

As mentioned above, we are currently repeating the key experiments with the [YL1/2] antibody, as suggested. Furthermore, we fully agree with the reviewer's point regarding polyglutamylation on β -tubulin. The C-terminal tail of β -tubulin does indeed contain polyglutamylation sites. As we noted in the manuscript (Lines 340-352), this aspect has not been investigated in the present study, and we acknowledge it as a valuable direction for future research. We will revise the text accordingly to avoid overinterpretation and to more accurately reflect the limitations of our current data.

<https://doi.org/10.7554/eLife.108209.1.sa0>